

Wired For Change? Clean Technology Adoption and Labor Market Transitions

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This version: June 7, 2026

Abstract

Whether the energy transition will harm or benefit workers is a central policy concern. This paper provides the first ex-post estimates of transition costs when small and medium enterprises adopt clean technologies. I study heating service firms in France as they shift from installing fossil fuel boilers to heat pumps. Tracking workers through matched employer-employee data, I document substantial within-firm labor reallocation but low transition costs overall. While displaced workers recover their earnings within one year, stayers adjust by working longer hours. Hourly wages rise for those exposed early to the technology or facing smaller skill gaps, but primarily through employer changes rather than within-firm raises. This demonstrates how market incentives can facilitate on-the-job skill updating, keeping adjustment costs low even amid significant labor reallocation.

Keywords: green jobs, energy transition, labor reallocation, skills upgrading

JEL codes: Q52, J24, O33

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Acknowledgements: This research is supported with funding from *Électricité de France (EDF)*. Access to some confidential data, on which this work is based, was made possible within a secure environment provided by CASD – Centre d'accès sécurisé aux données (Ref. 10.34724/CASD). I am grateful to Matthieu Glachant for his continuous advice and support. This paper greatly benefited from discussions with and comments by François Cohen, Misato Sato, and Aurélien Saussay.

1 Introduction

Climate policies raise significant concerns about job losses and reduced earnings, particularly for blue-collar workers in energy-intensive industries. These concerns fuel opposition to environmental regulations and contribute to anti-environmental political backlash¹. Green investment programs stand out as an important exception. By supporting the decarbonization of small and medium enterprises, they aim at facilitating within-occupation transition and supporting the creation of new green jobs². Yet success depends on whether workers can navigate this major transformation, and at what cost.

The energy transition demands skill updating across a wide range of occupations. Over 25% of all jobs will be significantly impacted by the net-zero transition, yet only 2.1% to 14.1% of training courses in developed economies deliver green content (OECD 2024), and demand for green skills already grows twice as fast as supply³. This training shortfall has not gone unnoticed: European policymakers estimate €1.7 to 4.1 billion will be needed for reskilling in net-zero manufacturing alone through 2030 (European Commission 2023). Despite substantial policy commitments⁴, actual adaptation costs borne by workers remain poorly understood. We lack systematic evidence on how workers—especially those in resource-constrained SMEs whose skills are most distant from new requirements—adjust when their employers adopt clean technologies.

This paper provides the first ex-post evidence of transition costs occurring within small and medium enterprises adopting clean energy technologies. I study French heating service firms transitioning from fossil-fuel boilers to heat pump installation. The shift is emblematic of SME-level green technical change as it implies major skill transformation. Beyond handling refrigerants and working safely with high-voltage circuits, heat pump requires deeper thermodynamic understanding than gas boilers, as pumps move heat through refrigeration cycles rather than burning fuel.

¹In the US, see: “Trump is still courting coal workers. This county shows why it matters.” 2024; “Reinvigorating America’s Beautiful Clean Coal Industry and Amending Executive Order 14241” 2025. In the EU, see “EU to pare back sustainability rules for companies, draft shows” 2025.

²Green recovery packages have come with massive job creation promises, potentially alleviating the negative consequences of the energy transition. Recent examples include the Inflation Reduction Act (IRA) in the US or the European Green Deal in the EU (European Commission 2020; The White House 2022).

³According to the *LinkedIn 2024 Global Green Skills Report*, green hires have grown 6.2% annually over 2021-2025, vs. 3.4% for green skills (LinkedIn Economic Graph 2024).

⁴Policy initiatives include the EU’s Just Transition Mechanism (2021) providing €55 billion for regions affected by decarbonization, the EU’s European Social Fund Plus (ESF+) for 2021–2027 aiming to retrain 5 million people for jobs in the green economy (European Commission 2021), the ILO’s Guidelines for a Just Transition (2015), and the G7 Green Jobs Initiative (2023) establishing international frameworks for measuring and supporting green skills development.

The French institutional context provides a unique setting for identifying technology adoption at the establishment level. As for any energy efficiency investments, confidence in contractors’ competencies is central concern for prospective customers. To address this challenge, an establishment-level quality certification has been created for each specific work-type. Since 2015, only works performed by certified contractors qualify for energy efficiency subsidies. The certification requires at least one worker to follow a five days training, both theoretical and practical. Trained workers are responsible for knowledge diffusion to their peers within firm, and become the establishment’s “environmental guarantor”. For heat pumps, training is focused on sizing calculations, performance optimization and system integration. The training also covers customer advisory and up-to-date knowledge of the regulatory framework, with a particular emphasis on product durability and environmental protection.

Crucially, the certification is designed as a first entry point into each specific market segment. Upon certification, contractors get listed on public registers, which makes it especially compelling for market entrants seeking credibility towards prospective customers. Certification thus provides an explicit marker of initial technology adoption, signaling the start of a competency-building process within the establishment. I use this marker to document a surge in heat pump installation proficiency starting in January 2019, when subsidies granted to households reached €2,500—equivalent to one fifth of the overall installation cost. This yields a quasi-exogenous surge in installation capability which I document using the number of certified establishments. Tracking workers through matched employer-employee data, I compare their labor market outcomes in adopting versus non-adopting establishments over 2015-2023. As heating firms switch to the new installation service, changes in earnings and career trajectories reflect the costs and benefits of adapting to this clean energy technology.

I find that technology adoption yields major labor reallocation at the establishment level. Using a staggered difference-in-difference estimation, I estimate 1.5 job creation and 0.75 job destruction within 2 years. Second, I find no evidence of lasting transition costs. In an event-study design tracking workers in the universe of heating service firms, I identify a sharp increase in both hours and labor earnings exceeding 10% in post adoption years. These results are driven by stayers, who do not separate from their pre-adoption employer and experience on-the-job re-skilling. Consistent with a high monopsony power of their employer, stayers benefit from very modest hourly wage increases. On the other side, workers who separate after adoption experience only short-run losses

and recover fully after one year. Hourly wages increase sharply for workers with prior technology exposure, consistent with rising market demand for heat pump installation expertise. Workers at non-adopting establishments also benefit joining, especially when they face smaller skill gaps.

These findings are relevant to a broad literature documenting the labor market consequences of environmental regulations. Concerns about a just transition are shaped by the case of workers in the fossil energy sector, especially coal-fired electricity generation and oil extraction⁵. Indeed, workers displaced from these polluting sectors, endowed with occupation-specific skills, incur major welfare losses (Walker 2013, Yip 2018, and Marin and Vona 2021). A growing literature focuses on the adverse employment effects of resource extraction declines, exemplified by the phase out of coal mining (Haywood, Janser, and Koch 2021, Rud et al. 2024) or the contraction of oil extraction (Ellingsen and Espegren 2022, Garnache, Isaksen, and Nareklshvili 2025). These earning losses are compounded by the fact that workers rarely succeed at moving from more pollution-intensive to greener jobs (Bluedorn et al. 2023, Curtis, O’Kane, and Park 2024).

This strand focuses predominantly on declining, hard-to-abate industries rather than on clean technology adoption in transitioning SMEs. A potential explanation is the empirical challenge to identify quasi-exogenous adoption of clean energy technology, a fundamentally endogenous decision with respect to each firm’s own strategy and transition costs. As a result, existing research cannot separate the effect of technology adoption at the establishment-level from that of the industry-level economic downturn. Yet effects may not generalize to SMEs, which gather 60% of workers (OECD 2023). While energy-intensive industries face fundamental technical and financial barriers⁶, the transition of SMEs may prove achievable through proven technologies requiring limited upfront investment (World Economic Forum 2024; European Investment Bank 2025). My empirical setting addresses this challenge by exploiting the unique context of energy efficiency policies in France. I leverage a subsidy shock and a legal requirement—the establishment-level environmental quality certification *RGE*—to identify a technological shift and evaluate its labor market effects. By isolating technology adoption from recessive dynamics, I observe both layoffs and hires, providing a much more nuanced understanding of the labor market effects of clean technology adoption.

The findings in this paper also contribute to the literature on skill-biased technological change

⁵Look et al. 2021 identify more than 100 US or European policies aimed at achieving “a widespread shift from coal to natural gas and renewables”, while “prioritizing fairness for workers and communities that have historically depended on fossil energy”.

⁶According to Gross 2021, an example of technical barrier is high-temperature heat requirements; financial barriers include capital intensity, long investment cycles and stranded assets.

in green transitions. The task-based framework developed by Autor, Levy, and Murnane 2003 and Acemoglu and Autor 2011 establishes that technological change reshapes labor demand by altering the task content of jobs. Gathmann and Schönberg 2010 demonstrate that workers move between occupations based on task similarity, with task-specific human capital accounting for up to half of wage growth. I validate these principles in the context of clean technology adoption: heat pump installation transforms task requirements within occupations, and adjustment costs vary systematically with skill distance.

Prior work documents the skill requirements of green jobs. Vona et al. 2015 and Vona et al. 2018 demonstrate that green occupations require more technical, engineering, and managerial skills. Consoli et al. 2016 confirm these differences persist when comparing green and non-green jobs within similar occupations. Saussay et al. 2022 advance this literature by using comprehensive job posting data, revealing that low-carbon vacancies demand higher skills across all five major skills groups—cognitive, IT, management, social, and technical. They document substantial heterogeneity across occupations, suggesting highly context-specific retraining needs.

I extend this literature in three ways. First, I document how workers actually acquire green skills: through on-the-job training at adopting establishments rather than through workforce replacement. Second, I show that skill premiums materialize primarily through job mobility, not within-firm raises—stayers gain 1% while job-switchers gain 20%. Third, I demonstrate that despite elevated skill requirements, workers successfully bridge skill gaps when transitions occur within related occupations, recovering earnings within one year. These results show that task distance and skill transferability are central to understanding adjustment costs in the energy transition.

The remainder of the paper is organized as follows. Section 2 provides background on France’s energy efficiency policy and the establishment-level certificate. Section 3 describes the data. Section 4 outlines the empirical strategy. Section 5 presents the results. Section 6 discusses the mechanisms and policy implications, and Section 7 concludes.

2 Context

As the host-country of the Paris Agreement of 2015, France has set ambitious mitigation targets. This includes a 30% reduction in gross greenhouse gas emissions by 2030 compared to 2012 lev-

els⁷. To achieve this goal, efforts concentrate on the decarbonization of the building stock, which represents 44% of the energy consumed and roughly one quarter of carbon emissions. According to the French Prime Minister’s strategic planning agency *France Stratégie*, 72% of all 2030-targets green investments should go to buildings energy renovation (Pisani-Ferry and Mahfouz 2023). This effort requires insulating buildings’ envelope and decarbonizing heating systems. In this paper, I focus on the second type of investment with the replacement of fossil boilers by heat pumps.

2.1 The French Energy Efficiency Policy

France’s energy efficiency policy relies on a mix of public and private subsidies, which largely benefits the residential sector. Both types of funding have changed in scale and scope since the mid-2000’s, when the first tax credit and the pilot period of the energy efficiency obligations scheme (EEOs) were launched. While public funding has transitioned to a means-tested subsidy mechanism in 2020 (France Stratégie 2024), the private sector remains involved through the EEOs. To fulfill their individual obligations, obligated parties support efficiency improvements at final consumers—mostly in the residential sector⁸. Each supported work generates energy saving certificates, which are used as proofs of compliance by obligated parties.

In September 2017, the French government announced a broad *Climate Plan*, which specifically targeted the phase out of fuel oil boilers and conventional (i.e., non-condensing) natural gas boilers⁹. The announcement marked an important milestone in the energy transition as fuel oil heating systems emit 324 grams of CO₂ per kWh, compared to 227 grams for natural gas boilers and only 49 grams for heat pumps (CGDD 2024). At the time in France, 3.9 million households (13%) relied on fuel oil and 11.9 million (41%) on natural gas as their main heating source (SDES 2022). Yet actual implementation of the plan was not immediate, and happened through a progressive evolution of the French energy efficiency policy framework over 2018-2020.

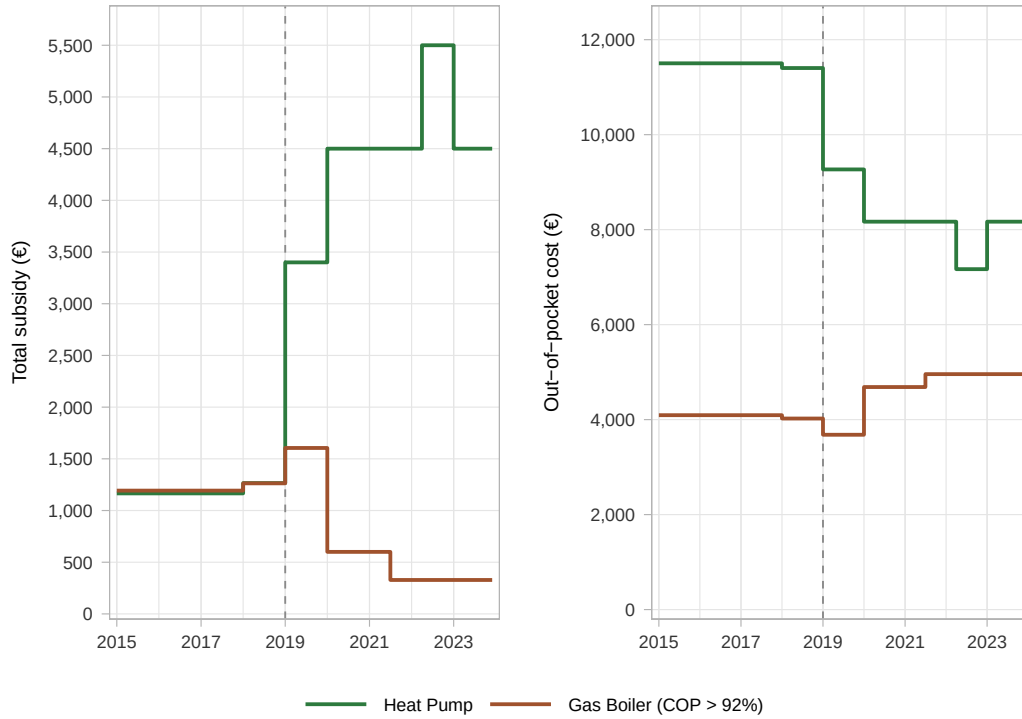
The fourth period of the French EEOs started in 2018. The total energy savings target for 2018-2020 was set at 1,600 cumulative TWh, or twice the amount of the previous period (2015-2017). Because of the higher compliance costs for obligated parties, the risk was that only affluent

⁷The 2024 *National Energy and Climate Plan* targets 107 million tonnes of oil equivalent of final energy consumption in 2030, which is 20% below the 2022 level (European Commission 2024).

⁸Each period-specific (3-4 years) national energy savings target breaks down into individual targets assigned to retailers of electricity, gas, and gasoline in proportion to their sales, with differing coefficients depending on the type of fuel and their carbon content. This is a direct application of the polluter pays principle (*TFEU* 2016).

⁹*Libération* on September 17, 2017 (“Le plan Hulot: quatre mesures écologiques et solidaires” 2017, *in French*).

Figure 1: Financial support and out-of-pocket costs for heat pumps & gas boilers (2015-2023)



(a) Financial support

(b) Out-of-pocket cost

Notes: EEOs grant + public subsidy at the median income (MaPrimeRénov' Violet). Installation costs from ADEME 2019: €12,668 (heat pumps), €5,286 (gas boilers). EEO grants from standardized fiches for 100 m², zone H1: 66,400 kWh cumac (heat pumps), 46,900 kWh cumac (gas boilers). EEO prices from Assemblée Nationale 2024: €4/MWh cumac (2015-17), €5.5/MWh (2018), €7/MWh (2019-23). Jan. 2019 *Coup de Pouce* (Ministère de la Transition Écologique 2018): €2,500 (heat pumps), €600 (gas boilers). CITE (2015-2018): €900 (heat pumps), €1,005 (gas boilers). In 2020, CITE rises to €2,000 for heat pump, while gas boilers are excluded. MaPrimeRénov' starts in 2021 with the same levels, up to €3,000 over Apr.-Dec. 2022 (Recover Plan). Gas boiler minimum EEO grants phased out June 2021.

household—able to pay for the remaining out-of-pocket cost—would benefit from renovation support. To increase political acceptance, the government introduced a bonus mechanism within the EEOs. Under this new system, suppliers supporting specific investment types could reduce their individual obligation if they committed to specific minimum grant levels (Darmais, Glachant, and Kahn 2024). In April 2018, minimum grants of €3,000 (first quartile) and €2,000 (second quartile) were created for households below the median income replacing fuel oil boilers by a heat pump. A few months latter in January 2019, minimum grants were extended to all income levels and conventional gas boiler replacement (i.e., any natural gas boiler except condensing ones). The subsidy for heat pumps rose to €4,000 for all households below the median income and €2,500 for households at or above the median income (Assemblée Nationale 2021). The reform also extended minimum grants to high-efficiency gas boilers, setting it to €1,200 for low-income households and

€600 for non-low income households. The subsidy shock was unprecedented by its size and its scope. As depicted in Figure 1, it created a fundamental shift in incentives, favoring households clean energy technology adoption.

The January 2019 reform marks the beginning of a sustained public support effort for heat-pump installations in France. In 2020, the tax credit was replaced with a direct, means-tested subsidy of €2,000 for households at the median income level. In contrast, high-efficiency gas boilers were removed from eligibility for public subsidies in 2020 and from minimum EEO grants in June 2021. Importantly, this shift in incentives does not imply that high-efficiency gas boilers are no longer a viable heating option. The right panel of Figure 1 illustrates the evolution of out-of-pocket costs for both technologies for median-income households. Throughout the period, gas boilers remain between three times cheaper and 40% cheaper than heat pumps. Nevertheless, the reform sends a strong signal in favor of decarbonized heating solutions to both households and installers.

2.2 Tracking establishment-level technology adoption

Energy efficiency investments are credence goods, making confidence in the quality of installation central to the development of the industry (Giraudet 2020). To ensure minimum quality standards and promote quality upgrading within the industry, policymakers created the “Reconnu Garant de l’Environnement (RGE)” certificate—literally, “Recognized Environmental Guarantor”. Since 2015, both public subsidies and energy supplier grants are conditional on hiring a certified contractor (Code général des impôts 2014). As depicted in Section A.2, each work type (insulation, heat pump installation, gas boilers, biomass systems, photovoltaic installations) requires a specific certificate. Appendix A.3 provides a detailed overview of the certification process, including training requirements, costs, and timeline.

2.2.1 The skill content of certification

Three features establish heat pump installation certification as a valuable knowledge input favoring technology adoption—rather than bureaucratic credentialing or windfall access to subsidies.

First, the heat pump certification aims at guaranteeing environmental performance outcomes—the core meaning of “Environmental Guarantor”. The five-day training covers three domains: (i) *Technical competencies* include codified knowledge about heat pump-specific thermodynamic

principles, sizing calculations, COP optimization, and system integration; (ii) *Regulatory and advisory competencies* cover up-to-date knowledge of subsidy eligibility, environmental regulations, and product durability standards; (iii) *Customer communication* requires explaining technology choices and long-term maintenance requirements to clients (Qualit’EnR 2024). These competencies are distinct from the tacit, hands-on skills typically transmitted through apprenticeship in traditional HVAC work.

Second, trained workers become technical referents within their establishment, transforming individual certification into organizational capability-building. They are responsible for transmitting knowledge to peers, establishing company-wide procedures, and maintaining quality standards (Service-Public.fr 2024).

Third, post-certification audits verify applied competence through performance-based evaluation. For heat pumps, classified as “critical” installations due to technical complexity, establishments face two independent audits within the certification cycle—one within 24 months of certification and another in year N+1 (Ministère de la Transition Écologique 2021). These audits assess actual installation performance against environmental standards rather than theoretical knowledge acquisition.

2.2.2 Certification as a credible sign of technology adoption

Subsidy coverage patterns validate RGE certification as capturing genuine technology adoption rather than windfall credentialing of pre-existing capabilities. Indeed, tax authority data reveal substantial coverage of the heat pump renovation market by certified firms in the pre-2019 period. Over 2016-2018, tax credit beneficiaries ranged from 23 to 38K households annually, while EEO grants supported 5-9K installations. Crucially, both the tax credit and EEO grants required RGE-certified contractors. This means at least two thirds of all renovation installations were performed by certified firms, who are also more likely than non-certified firms to perform any heat pump installations. This demonstrates certification was economically necessary for market participation, making it a reliable proxy for technology adoption. Detailed computation regarding this substantial coverage of the renovation market are detailed [Appendix A.4](#).

The certification program is explicitly designed as a market entry mechanism for each technology segment. The French energy agency (ADEME) positions RGE qualifications as enabling contractors to “develop their activity in the energy renovation market” (ADEME 2019, p. 8),

while the certification body Qualit'EnR emphasizes that certification allows professionals to "position themselves on the renewable energy equipment market" (Service-Public.fr 2024). Upon certification, contractors are listed on public registers (*annuaire des professionnels RGE*), which government agencies actively promote to households seeking subsidized renovations (France Rénov' 2024). This institutional design makes certification especially compelling for market entrants seeking credibility with prospective customers. For heating service establishments, obtaining heat pump certification thus provides an explicit marker of initial technology adoption, signaling the start of a competency-building process within the establishment.

Figure 2 presents establishment-level certification patterns following the January 2019 subsidy increase. Certification rises from 5% of heating service establishments in late 2018 to 9% by mid-2020, representing approximately 4,000 newly certified establishments—a 40% expansion of the certified base.

Figure 2: Cumulative Adoption Rate



Notes: The figure shows the supply-side response to the increase in heat pump subsidies over 2019 (dashed lines). The left panel displays the cumulative share of heating service establishments certified to install subsidized heat pumps in France. The right panel shows monthly certification counts.

For the 2,859 establishments adopting in 2019, entry into the subsidized heat pump market represented genuine technology adoption. In 2018, only 5 percent of the approximately 100,000 heating service establishments in France held heat pump RGE certification, underscoring the scale

of the competency gap that the 2019 subsidy shock set in motion. The near-universal subsidy coverage pre-2019 ensured that heat pump-capable establishments certified regardless of subsidy levels, making the 5% baseline a revealed upper bound on pre-existing capabilities within the heating service sector. The 2019 subsidy shock—extending *Coup de Pouch Chauffage* to all households and conventional gas boiler replacements and raising grants to € 2,500—creates an unprecedented incentive for heating service firms to enter the subsidized heat pump market. This exogenous shock isolates technology adoption from firm fundamentals, offering a unique opportunity to causally estimate the labor market impacts of clean energy technology adoption at the establishment level.

3 Data

I combine several types of administrative datasets to construct the complete universe of heating services businesses in France and their workforce dynamics around the time of the reform.

The analysis primarily draws on employer-employee linked datasets. The first one is the *Mouvements de Main d’Œuvre (MMO)* dataset (DARES 2025) that provides daily records of all employees entries and exits from any establishment in France, including detailed information on contract types, reasons for exits (economic layoffs, retirement, etc.), and socio-professional categories. The second one is the Base Tous Salariés (INSEE 2025a), which provides yearly hours worked and labor earnings for each wage earner in each business in France. This exhaustive administrative dataset is derived from employer tax declarations. A key feature of these data is the construction of a quasi-exhaustive longitudinal panel, which required overcoming the challenge that individuals are assigned new pseudonymous identifiers each year. Babet, Godechot, and Palladino 2025 developed a matching algorithm that exploits overlapping annual files to track workers across time, achieving approximately 98% successful matching for 2002-2023. This “wide panel” represents a substantial improvement over the traditional narrow panel (1/12th sample) previously used for French labor market analysis, enabling more precise estimation of workers labor market outcomes by capturing the full universe of mobile workers (see Appendix B.1 for detailed methodology).

To identify heat pumps technology adoption, I use the RGE (Reconnu Garant de l’Environnement) registry (ADEME 2025) from the French Energy Management Agency. This certification system indicates whether an establishment is qualified to install subsidized energy-efficient equipment. Each type of work requires specific certification, enabling precise identification of the timing of

technology adoption by any certified establishment. I also combine it with the registry for works supported by the French Energy Saving Obligation (*Certificats d'Économie d'Énergie*, *CEE* in French).

Finally, I rely on the SIRENE database (INSEE 2025b), which serves as the national comprehensive directory for identifying French companies and their establishments. It includes the main activity code (APE at the 5-digit level, in 732 sub-classes), the exact geographic coordinates, and the date of creation of each establishment.

3.1 Defining treatment and control groups

A general concern when studying the labor market effects of technology adoption is the risk of spillovers from treated to untreated establishments and workers (the so-called *stable unit treatment value assumption* — *SUTVA*). Indeed, heating service firms operate on local markets, and any treatment effect of technology adoption could result in competition or equilibrium dynamics, contaminating the outcomes of seemingly untreated establishments and their workers. On the other hand, heating system installers are exposed to specific dynamics, such as the broad political momentum that benefited energy efficiency investments (heat pumps, but also energy efficient gas boilers) at the French and EU level over 2019-2022¹⁰. This major stimulus should not conflate the effect of technology adoption itself which restricts the choice of potential control establishments.

These two objectives create a tradeoff between avoiding heat pump adoption spillovers and maintaining comparability within the heating system installation industry. To address this challenge, I exclude establishments that adopt heat pump certification after 2019, referred to as “not-yet-treated” units in the modern DiD literature (Chaisemartin and D’Haultfœuille 2020). The exclusion strategy leverages the daily register of heat pump certifications through 2023 to identify future adopters. The underlying rationale is that substantial spillovers should trigger non-adopters facing competitive pressure from early adopters to adopt themselves the certification after 2019. This is supported by the low cost of certification (between €500 and €1,500) and is further reinforced by two contemporaneous developments during the study period. First, the COVID-19 pandemic accelerated the availability of online training programs, substantially reducing practi-

¹⁰In 2019, France enacted its *Energy and Climate Act*, which legislated a net zero emissions target for 2050. The law introduced several obligations attached to poorly insulated homes, such as mandatory energy audits before renting or selling, and a progressive ban on rental properties below a set level of energy efficiency. This is in line with the EU’s *Renovation Wave Strategy* announced in 2020, which aims to renovate 35 million buildings by 2030, at least doubling the annual rate of energy renovations in the EU.

cal barriers to certification (FEEBAT 2020). It dramatically reduced travel and accommodation expenses related to the certification, allowing professionals to continue managing their business activities through the training period. Second, the post-pandemic recovery package included a dramatic expansion of subsidies for energy-efficient heating systems, creating strong incentives for establishment-level adoption. Consequently, establishments remaining uncertified through 2023 despite the low adoption costs and the substantial policy incentives likely experienced very few competitive pressures from others, adopting firms. In practice, as shown in section 3.2, control establishments face different markets less favorable to heat pumps adoption, characterized by a lower reliance on fuel oil heating¹¹, a higher share of multifamily dwellings, or less affluent customers for whom the out-of-pocket cost remained too high even with the increased subsidies. Never-treated establishments therefore provide a valid control group, largely insulated from local spillovers and equilibrium effects.

3.2 Samples description

3.2.1 Establishment-level sample.

The sample construction follows a cross-referencing identification process. First, I identify establishments with heat pumps RGE certificates and extract the list of main activity codes (APE at the 5-digit level). Figure 15 shows the top 10 main activity codes. Second, to distinguish systematic sectoral presence from occasional diversification, I retain only APE codes representing at least 5% of all certified establishments¹². The integration of these datasets creates a complete universe of heating service businesses in France from 2015 to 2023. Finally, I keep establishments that are either never certified, or that certified for heat pump installation in 2019. This yields a sample of 95,418 active establishments.

I aggregate the establishment-level employment records and create a quarterly panel spanning from 2017 to 2023. Table 1 presents establishment-level characteristics for treated and control groups in the pre-policy period (2017-2018). Average headcount is nearly identical at approximately 6 employees, and establishment age is also well-balanced at around 7 years, indicating that treated and control establishments are comparable in terms of size and maturity.

¹¹Figure 12 maps the spatial heterogeneity in fuel oil reliance pre-policy.

¹²This includes heating and HVAC¹³ (43.22B), water and gas (43.22A), and electrical (43.21A) installation.

Table 1: Pre-treatment balance (Establishment level, 2017-2018)

	Treated		Control		Difference (T-C) (5)
	Mean (1)	SE (2)	Mean (3)	SE (4)	
N Establishments	2,859		92,559		
Headcount	6.07	(12.05)	5.80	(19.37)	0.27
Entries	2.75	(6.90)	2.49	(6.60)	0.26
Exits	1.21	(2.76)	1.03	(2.58)	0.18
Turnover rate	0.700	(1.11)	0.82	(3.92)	−0.12
Age (years)	6.71	(9.81)	6.92	(9.27)	−0.22
Population (CZ)	20,577	(48,058)	34,199	(60,580)	−13,323
Share Fuel Oil (CZ)	11.6	(8.4)	8.9	(7.4)	2.6

Notes: The table shows summary statistics for treated and control establishments in the pre-policy period (2017-2018). Treated establishments are those that adopt heat pump certification during the study period (2017-2021). Control establishments never adopt. Standard errors in columns (2) and (4). All statistics calculated as establishment-level averages over 2017-2018.

Three main differences emerge between the groups. First, treated establishments are located in less populated areas, with an average population of 20,577 inhabitants in their commuting zone (CZ) compared to 34,199 for control establishments. Second, treated establishments operate in CZ with an average reliance on fuel oil of 11.6%. This is 2.6pp, or 30% higher than in controls' CZ. Third, control establishments exhibit higher baseline labor market turnover, with a rate of 0.82 compared to 0.7 for treated establishments. These differences point to the same direction: treated establishments operate in less populated, more rural markets where the share of detached housing units equipped with an old fossil fuel boiler is higher. The result is a differential incentive regarding heat pump installation certification, which supports the validity of the never-treated group as a credible counterfactual.

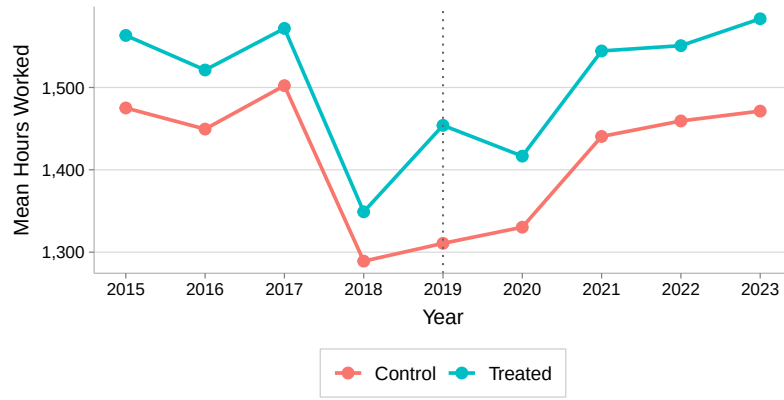
3.2.2 Worker-level sample.

I create a yearly panel gathering 800,000+ individuals who eventually worked in an establishment of the heating service industry across 2015 to 2023. This dataset provides yearly labor earnings and hours worked, as well as the unique identifier of their main employer’s establishment. Socio-economic variables include age and gender, as well as occupational classification assigned by employers following the comprehensive French occupational classification system (*Catégories Socioprofessionnelles*, *CSP*). All workers are employed full-time (at least 30 hours per week). The register includes all observations over 2015-2023, including for periods of employment outside the heating service industry.

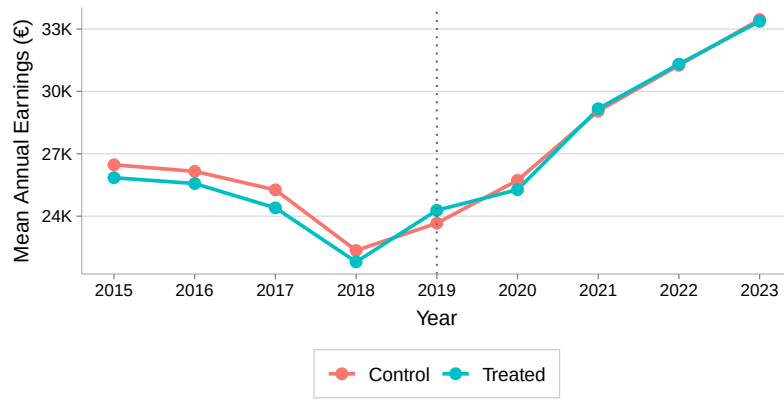
I define treatment as being employed in 2018 and 2019 in an establishment that adopts heat pump technology in 2019. To construct control groups, I match each treated worker to control workers at non-adopting establishments using exact matching on 2018 establishment activity code (APE), socio-professional category (CSP), and gender, followed by nearest-neighbor matching on age. I implement 1:20 matching, pairing each treated worker with up to 20 control workers. This n-to-many matching allows me to leverage the full potential of my sample of 800,000+ worker trajectories. The matching procedure yields 13,499 treated workers (68% retention from 19,921) and 121,681 matched controls (from 479,762). The 32% attrition reflects treated workers for whom no exact match exists across all three dimensions. Nearest-neighbor matching on age achieves near-perfect balance, with treated and control workers at 36.7 and 36.8 years respectively in the matched sample. For gender and occupational composition, aggregate balance is not expected as exact matching controls for sample-level differences within strata rather than through achieving overall balance. The validity of the identification strategy thus relies on parallel pre-treatment trends in outcomes. Appendix B.3 provides detailed balance statistics across all covariates.

Figure 3 displays the evolution of hours worked, annual earnings, and hourly wages for matched treated and control workers from 2015 to 2023. In 2018, hourly wages averaged €15.45 for treated workers and €16.46 for controls. These wages are slightly above the French median hourly gross wage of €15.2 but remain well below the national mean of €18.1 (INSEE 2021). Workers earn approximately 1.5-1.7 times the minimum wage (€9.88 per hour in 2018). Annual earnings of approximately €22,000 reflect both wage levels and hours worked (around 1,300-1,400 hours annually), substantially below the national average gross annual salary of €36,238, consistent with the blue-collar composition of the workforce. The wage gap between treated and control workers

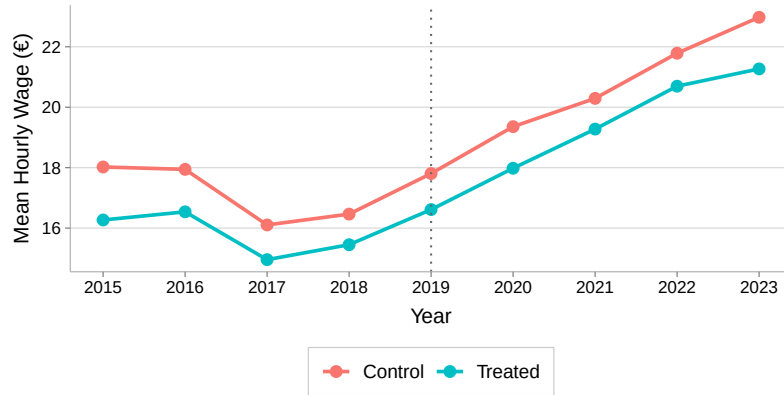
Figure 3: Worker Outcomes Over Time in the Matched Sample (2015-2023)



(a) Hours worked



(b) Annual earnings



(c) Hourly wage

Notes: The figure shows the evolution of hours worked, annual earnings, and hourly wages for matched treated and control workers from 2015 to 2023. Treated workers are employed at establishments that adopt heat pump technology in 2019 (vertical dashed line). Control workers are matched using exact matching on establishment activity code (APE), socio-professional category (CSP), and gender, followed by 1:20 nearest-neighbor matching on age. The parallel trends in all three outcomes during the pre-treatment period (2015-2018) support the identifying assumption for the difference-in-differences analysis.

(€1.01 per hour in 2018) reflects compositional differences documented at the establishment level: treated workers are concentrated in more rural areas with lower wage levels.

Critically, Figure 3 demonstrates parallel trends in all three outcome variables during the pre-treatment period (2015-2018). This supports the identifying assumption, setting the stage for the difference-in-differences estimation strategy presented in Section 4.

4 Empirical Strategy

To investigate the labor market outcomes of technology adoption, I exploit the surge in heat pumps installation subsidies and the subsequent response of local energy services contractors. My identification strategy leverages a key institutional feature: the requirement for installers to obtain RGE (Reconnu Garant de l'Environnement) certification to access the subsidized market. The certification process involves training workers in heat pump installation techniques and meeting quality standards. The price of the training is rather cheap, with an average cost of €500 per establishment. Upon certification, installers can access the market for subsidized installations. I use two distinct strategies to document the causal effect of technology adoption. First, I focus on employment policy at the establishment level. Second, I analyze earnings and hours at the individual level.

4.1 Establishment-level analysis

The policy-driven adoption of heat pumps certification by heating service establishments exhibits a staggered timing, as shown by the total adoption curve in Figure 2. I leverage this behavior and the quarterly frequency of my establishment-level dataset to estimate the effect of technology diffusion on employment policy. Staggered treatments pose several econometric challenges, and recent methodological advances have highlighted potential biases in traditional two-way fixed effects estimators when treatment effects are heterogeneous and treatment timing varies (Goodman-Bacon 2021; Callaway and Sant'Anna 2021). In this context, treatment intensity likely varies across establishments, with rural and smaller firms potentially experiencing larger effects due to their own and market characteristics. More importantly, early adopters should not be used as control units for late adopters as their treatment effect is not stable across time, which might bias the estimation. To address these concerns, I employ the Callaway and Sant'Anna 2021 methodology,

which provides several advantages for my setting. First, I use never-treated establishments to define a clean control group, avoiding contamination from already-treated units. Second, I estimate group-time specific effects $ATT(g, t)$ for each cohort g (defined by certification timing) and time period t .

For each establishment i , cohort g and period t , I estimate:

$$ATT(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g] \quad (1)$$

where $Y_{it}(g)$ represents the potential outcome under treatment timing g , $Y_{it}(\infty)$ represents the never-treated potential outcome, and $G_i = g$ indicates establishments first certified in period g . This approach allows me to examine dynamic treatment effects while avoiding the pitfalls of conventional difference-in-differences estimators in settings with staggered adoption and heterogeneous effects. I apply this estimation strategy to four establishment-level outcomes. First, I document technology substitution through the share of air-to-water heat pumps and fossil fuel boilers among all subsidized installations. Second, I track entries and exits separately.

4.2 Worker-level analysis

To estimate the causal effect of heat pump adoption on worker outcomes, I rely on the matched sample of workers at adopting and non-adopting heating service establishments detailed in Section 3.2.2. The matching procedure addresses potential compositional differences between workers at establishments that adopt heat pump technology in 2019 and those at non-adopting establishments. Since treatment occurs at the establishment level through the adoption decision, worker selection into treatment is not a primary concern. Rather, matching ensures that treated and control workers are comparable in terms of observable characteristics that may influence labor market trajectories independently of the technology shock.

The worker-level analysis proceeds in three parts, each focusing on a distinct subset of workers. First, I estimate the main effect of heat pump adoption on all workers employed at treated establishments in both 2018 and 2019 (to exclude newcomers joining post-adoption in 2019). Second, I decompose treatment effects by distinguishing between stayers and leavers to understand compositional changes within adopting establishments. Stayers are workers employed at the same establishment in both 2018 (pre-shock) and 2023 (the final year of the panel), while leavers are

workers present at the establishment in 2018 or 2019 but no longer employed there in 2023. These two groups form complementary subsets of the treated workforce, enabling a comparison of outcomes for workers who remain versus those who separate. Third, I examine the labor market trajectories of movers, comprising both leavers and newcomers, in relative time. I define newcomers as workers not employed at a treated establishment in 2018 or 2019, but present in 2023, having entered the establishment in 2020 or later. For each analysis, I construct a new matched sample specific to the subgroup of treated workers under study, ensuring that treatment and control groups remain comparable within each estimation framework.

4.2.1 Dynamic difference-in-difference of heat pump adoption

The shift from gas boilers to heat pumps affects workers employed at treated establishments in 2019. I estimate the effect of technology adoption on worker outcomes using an event study specification. The baseline model is:

$$\log(y_{ist}) = \sum_{k \neq -1} \beta_k \cdot \mathbb{1}\{t = k\} \cdot \text{Treat}_{st} + \alpha_i + \gamma_s + \delta_t + \varepsilon_{ist} \quad (2)$$

where y_{ist} is hours worked or labor earnings for worker i in establishment s in year t , Treat_{st} is an indicator for establishment s ever adopting heat pump technology, and k measures years relative to 2019 (the adoption year). I include worker, establishment, and year fixed effects (α_i , γ_s , δ_t), and normalize 2018 ($k = -1$) as the reference period. The coefficients β_k trace out the dynamic treatment effects. Standard errors are two-way clustered at the worker and establishment levels.

4.2.2 Stayers vs. leavers: Calendar time comparison

To account for workforce composition effects, I decompose treatment effects for stayers versus leavers, both expressed in calendar time. For stayers (workers present at their 2018 employer through 2023), I estimate equation 2 on the restricted sample of treated workers. For leavers, following the job displacement literature (e.g., Schmieder, Wachter, and Heining 2023), I focus on workers with at least three years of tenure at their pre-separation employer. I estimate the following specification that allows for cohort-specific treatment effects:

$$\log(y_{itc}) = \sum_{k \neq -1} \beta_k^c \times \mathbb{1}\{t = c_i + k\} \times \text{Leaver}_i + \alpha_i + \delta_t + \varepsilon_{itc} \quad (3)$$

where c_i denotes the calendar year when worker i separates (exit cohort), k is years relative to separation, and Leaver_i indicates treatment status compared to matched controls. The interaction $\mathbb{1}\{t = c_i + k\}$ links relative time to calendar years, yielding cohort-specific coefficients β_k^c . I then aggregate these coefficients to calendar time by computing the average effect across all cohorts active in each year:

$$\gamma_t = \frac{1}{N_t} \sum_{\substack{c,k: \\ c+k=t}} \beta_k^c \quad (4)$$

where N_t is the number of cohort-relative time pairs contributing to year t . Standard errors for γ_t are computed using the variance-covariance matrix of the β_k^c estimates to properly account for covariance across the averaged coefficients. This approach, detailed in section C.1, allows direct comparison of treatment effects on stayers versus leavers in the same calendar-time framework.

4.2.3 Movers: Relative time dynamics

To understand the dynamics of worker mobility around the technology transition, I estimate relative-time event studies separately for leavers and newcomers. Leavers are workers with at least three years of tenure who separate from their 2019 employer; newcomers are workers not employed at a treated establishment in 2019 but present in 2023. For both groups, I estimate:

$$\log(y_{ist}) = \sum_{k \neq k_0} \beta_k \times \mathbb{1}\{t = T_i + k\} \times \text{Mover}_i + \alpha_i + \gamma_s + \delta_{z \times t} + \varepsilon_{ist} \quad (5)$$

where T_i is the year of the mobility event (separation for leavers, entry for newcomers), Mover_i indicates treatment status, γ_s are establishment fixed effects, and $\delta_{z \times t}$ are employment zone-by-year fixed effects that flexibly control for local labor market conditions. The reference period is $k_0 = -1$ for leavers (the year before separation) and $k_0 = -2$ for newcomers (two years before entry, as workers entering at $t = 0$ likely separate from their previous employer at $t = -1$). Both leavers and newcomers are compared to matched control workers selected using exact matching on age, socio-professional category (PCS), establishment main activity code, and gender. Standard errors are two-way clustered at the establishment and worker levels.

5 Results

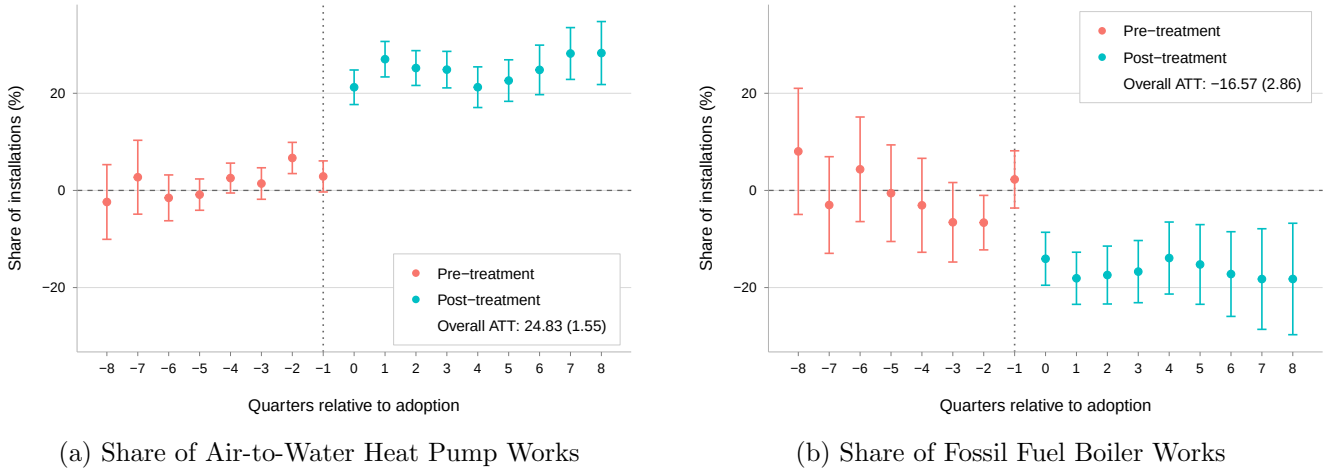
5.1 Effects on establishments employment behavior

I estimate the staggered difference-in-differences introduced in equation 1 on each quarter from 2017 to 2023. Averaging daily variables on quarters smooths idiosyncratic variation from high-frequency installation and employment records while reducing computational burden. I apply this estimation strategy to four establishment-level outcomes: the share of heat pump and fossil fuel boiler installations, job entries and exits.

5.1.1 Technology substitution

Figure 4 establishes that RGE certification translates into genuine technology substitution at the establishment level. Following certification, treated establishments increase the share of heat pump installations by approximately 20 percentage points within two years (panel 4a). Simultaneously, the share of fossil fuel boiler works declines by approximately 20 percentage points over the same horizon (panel 4b).

Figure 4: Technology Substitution Following Heat Pump Certification



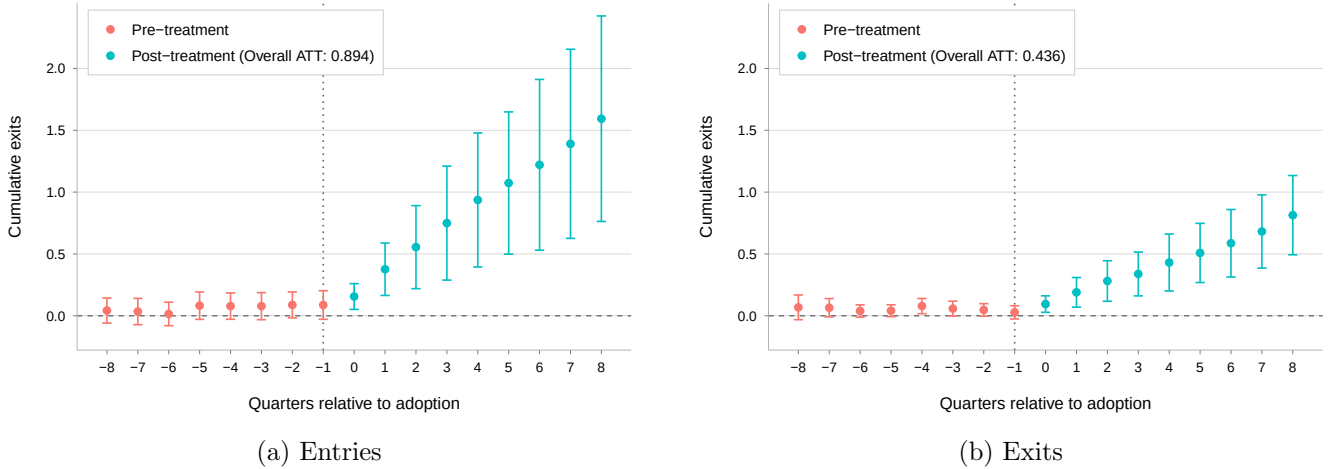
Notes: Event-study estimates of heat pump certification effects on establishment-level technology mix using Callaway and Sant’Anna 2021. Panel (a): share of air-to-water heat pump installation. Panel (b): share of fossil fuel boiler installation. Sample: 2,859 treated and 92,559 never-treated control establishments, 2017–2023 (quarterly). Dynamic aggregation with 95% confidence intervals. Standard errors clustered at establishment level.

Pre-treatment estimates are statistically indistinguishable from zero in both panels, supporting the parallel trends assumption. The near-perfect symmetry between the rise in heat pump activity and the decline in fossil fuel boiler work confirms that certification marks genuine product substitution within the establishment rather than a mere expansion of service scope. This is consistent with the institutional design of the RGE program as a market entry mechanism for a specific technology segment: adopting establishments do not simply add heat pumps to their portfolio—they restructure their activity around the new technology.

5.1.2 Employment: job creation and destruction

Figure 5 presents the cumulative employment effects, showing that 2 years after adoption, the average establishment experiences both higher job creation and higher job destruction.

Figure 5: Job Entries and Exits Following Heat Pump Adoption



Notes: Event-study estimates of heat pump certification effects on cumulative entries (a) and exits (b) using Callaway and Sant’Anna 2021. Sample: 2,859 treated and 92,559 never-treated control establishments, 2017–2023 (quarterly). Dynamic aggregation with 95% confidence intervals. Standard errors clustered at establishment level.

The pre-treatment estimates are statistically indistinguishable from zero, supporting the parallel trends assumption. Employment effects emerge immediately upon certification: adopting establishments experience approximately +1.5 job creation and +0.75 job destruction 2 years after certification. These are sizable effects relative to the average headcount of 6 employees.

While positive net job creation is consistent with expanding demand for heat pump services, the substantial job destruction occurring simultaneously is more concerning. The magnitude of worker

separations, which represents approximately half the rate of new hires, suggests that technology adoption triggers significant labor reallocation rather than simple workforce expansion, creating potential winners and losers among incumbent workers. Whether displaced workers successfully transition to new opportunities or bear substantial adjustment costs, and how these costs compare to gains for newly hired workers, cannot be assessed at the establishment level. I turn to individual worker trajectories in Section 5.2 to quantify these worker-level labor market effects.

5.2 Workers-level Career Trajectories

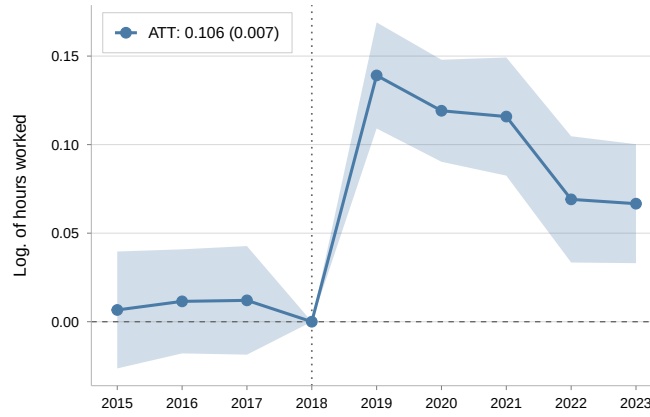
5.2.1 Effect on incumbent workers

I start by estimating the effect of heat pump certification on all workers employed at treated establishments in 2018 and 2019, regardless of whether they subsequently remain at or separate from the establishment. This provides an overall assessment of exposure to technology adoption for the incumbent workforce. I construct a matched control group of workers at never-treated establishments as detailed in section 3.2.2. I then estimate standard dynamic difference-in-differences (DiD) specifications comparing outcomes for treated and control workers over 2015-2023. Figure 6 presents the results for three labor market outcomes: log. total hours worked, log. annual labor earnings, and log. hourly wages.

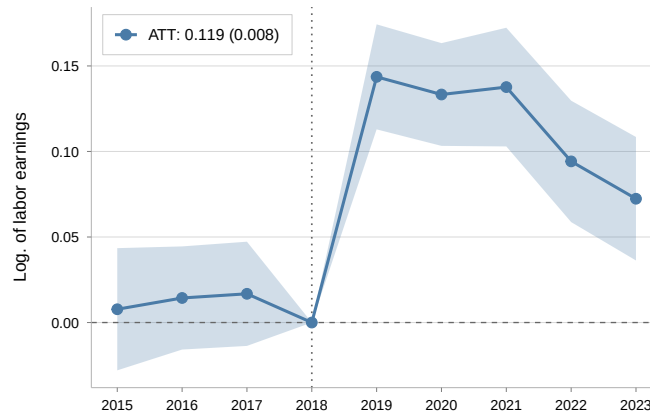
Several patterns emerge. First, treated workers experience substantial increases in both hours worked and labor earnings beginning immediately in 2019, the year of certification adoption. Hours worked and earnings increase by approximately 14 percent relative to matched control workers. These effects slightly decrease in subsequent years, but the average treatment over the treated (ATT) remains at or above 10 percent by 2020-2021. Second, the dynamics for hourly wages differ markedly from hours and earnings. Hourly wages show no significant effect in 2019 despite the large increases in hours and earnings. Significant hourly wage gains emerge only in 2020-2022, reaching approximately 2 percent above control workers. This pattern indicates that the primary adjustment margin in the immediate aftermath of certification is labor supply: workers increase hours at roughly constant hourly wages, with wage adjustments occurring only in subsequent years.

These average effects, while positive overall, mask substantial heterogeneity in outcomes between workers who remain at adopting establishments and those who separate. I turn to this decomposition in the next subsection.

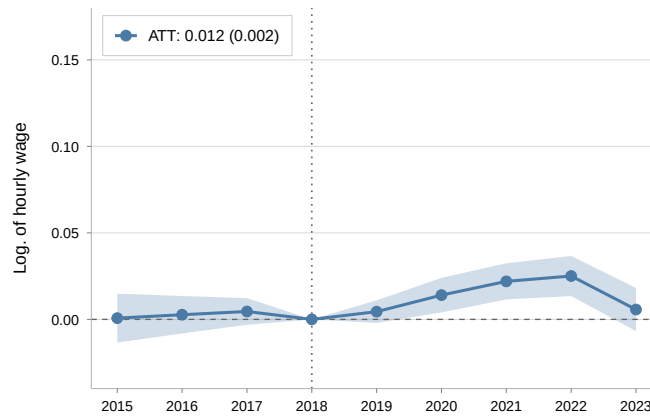
Figure 6: Effects of Heat Pump Certification on Incumbent Workers



(a) Log. Total Hours Worked



(b) Log. Annual Labor Earnings



(c) Log Hourly Wage

Notes: Event-study estimates of the effect of heat pump certification on labor market outcomes for workers employed at treated establishments in 2018 and 2019. The sample includes 11,229 treated workers and 117,748 matched control workers from never-treated establishments. Control workers matched exactly on establishment activity code (APE), socio-professional category (CSP), and gender, followed by nearest-neighbor matching on age. Shaded areas represent 95% confidence intervals based on standard errors clustered at the establishment level. Post-treatment ATTs in the legend are precision-weighted averages over 2019-2023.

5.2.2 Decomposing Effects: Stayers versus leavers

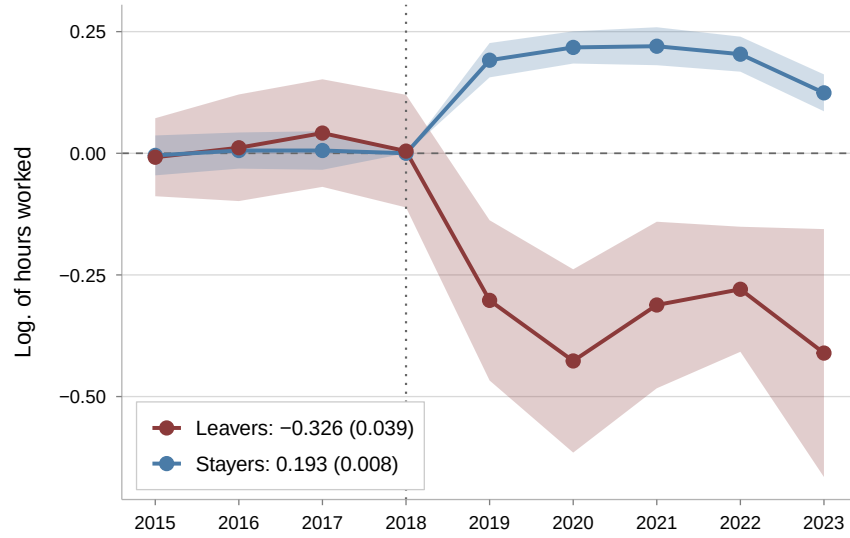
The average positive effects documented for incumbent workers aggregate over two distinct groups: workers who remain at adopting establishments (stayers) and those who eventually separate (leavers). To understand the composition of these aggregate effects, I decompose treatment effects by employment continuity. I define stayers as workers employed at the same establishment in both 2018 (pre-adoption), 2019 (treatment year) and 2023 (the final year of the panel); and leavers as workers present in 2018 and 2019 but no longer employed at the establishment by 2023. As a result, the leavers group aggregates in one sample all cohorts of exits over 2020-2023.

For both groups, I construct a new matched control sample using the same matching procedure detailed in section 3.2.2, which yields 7,395 treated stayers matched to 100,890 controls. Following the job displacement literature (e.g., Schmieder, Wachter, and Heining 2023), I further restrict the treatment group to workers with at least three years of tenure at their pre-separation employer. This yields 2,523 treated leavers matched to 36,576 controls. Figure 7 present the results for hours worked and labor earnings. The contrast between stayers and leavers is stark. Stayers experience immediate and substantial gains beginning in 2019, with hours worked and labor earnings increasing by approximately 20 percent relative to matched controls. These gains persist through 2022 before moderating slightly to 12.5 percent in 2023. In sharp contrast, leavers experience deteriorating outcomes following certification adoption. Their hours and earnings decline progressively, stabilizing around -30 percent by 2021-2022 relative to matched control workers who also eventually separate from their establishments.

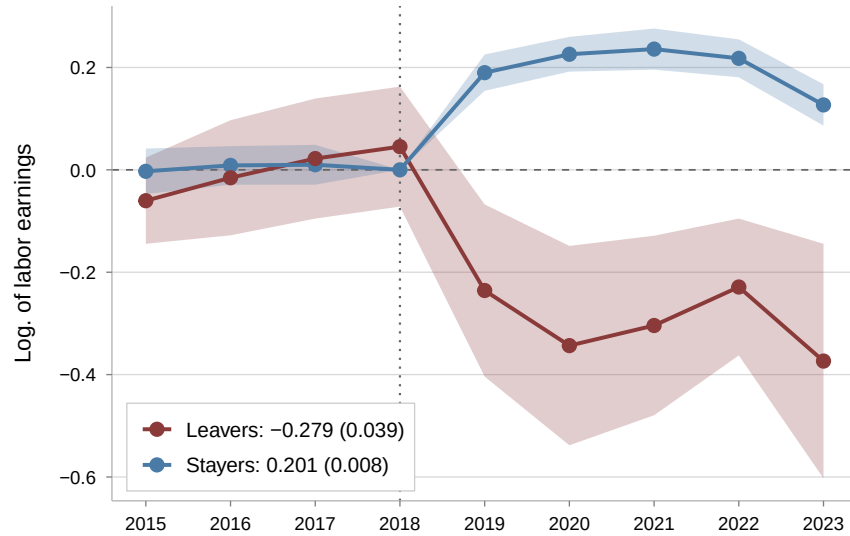
For stayers, I can additionally examine hourly wages. Figure 8 shows that stayers experience modest wage gains of around 1 percent beginning in 2020, consistent with the pattern observed for incumbent workers overall, although slightly lower.

The combination of increased hours and weakly higher wages explains the substantial earnings gains observed for stayers. The negative outcomes for leavers is consistent with substantial losses from separation. However, this interpretation requires careful consideration. The leaver sample in any given calendar year comprises workers who separated at different times following adoption, some immediately in 2019, others in 2020, 2021, or later. Consequently, the calendar-time effects aggregate over workers at different stages of their post-separation trajectories. A worker who separated in 2019 and found new employment appears in the 2022 estimate alongside workers separating for the first time in 2022. This aggregation obscures individual adjustment dynamics

Figure 7: Effects of Heat Pump Certification on Stayers versus Leavers



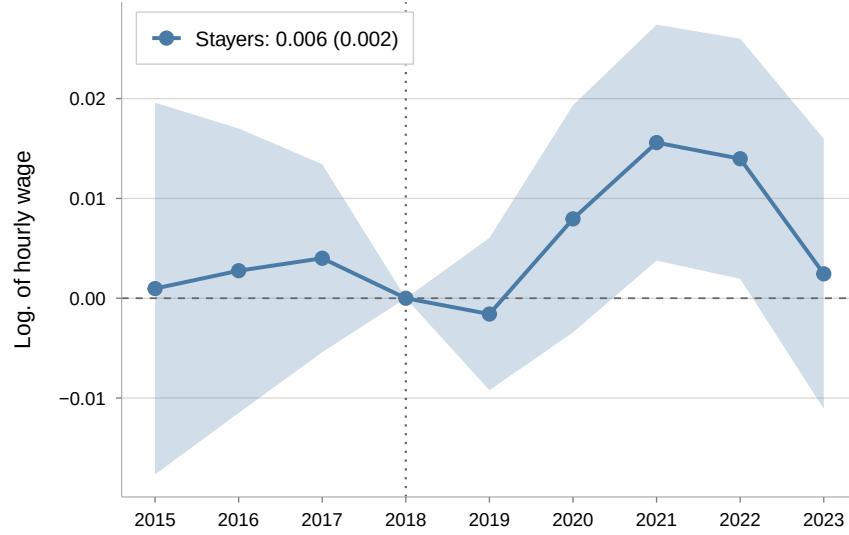
(a) Log. Total Hours Worked



(b) Log. Annual Labor Earnings

Notes: Event-study estimates of the effect of heat pump certification decomposed by employment continuity. Stayers (blue lines) are workers employed at the same establishment in both 2018, 2019, and 2023 ($N = 7,395$ treated, 100,890 matched controls). Leavers (red lines) are workers present in 2018 and 2019 but no longer employed at the establishment by 2023, having separated between 2020 and 2023 ($N = 2,523$ treated, 36,576 matched controls). Each treatment group is matched to its own control group of workers at never-treated establishments using the procedure detailed in section 3.2.2. Shaded areas represent 95% confidence intervals based on standard errors clustered at the establishment level. Post-treatment ATTs in legend are precision-weighted averages over 2019-2023.

Figure 8: Effects of Heat Pump Certification on Hourly Wages: Stayers



Notes: Event-study estimates of the effect of heat pump certification on log hourly wages for stayers—workers employed at the same establishment in both 2018, 2019 and 2023 ($N = 7,395$ treated, 100,890 matched controls). Hourly wage estimates for stayers only, as leavers experience unemployment following separation, precluding meaningful calendar-time comparisons across cohorts. Matching procedure and specification identical to Figure 7. Shaded area represents 95% confidence intervals based on standard errors clustered at the establishment level. Post-treatment ATTs in legend are precision-weighted averages over 2019-2023.

and may not reflect the true costs borne by displaced workers. Understanding whether movers actually get persistent individual-level losses or only incur a temporary transition requires tracking workers in event time relative to their displacement. I turn to this analysis in the next subsection.

5.2.3 Worker Mobility and Adjustment Dynamics

The previous analysis revealed that workers who separate from adopting establishments experience deteriorating outcomes in calendar time, but this comparison aggregates workers at different stages of their post-displacement trajectories. To assess individual-level adjustment costs, I examine worker outcomes in event time relative to their mobility event. I focus on two groups of movers: leavers who separate from treated establishments, and newcomers who enter them. While both groups involve labor mobility, they represent distinct phenomena. Leavers exit establishments that adopted certification in 2019, while newcomers enter these establishments after adoption.

For leavers, I use the definition introduced in subsection 5.2.2, which requires workers to be at the same treated establishment from 2016 to 2019 onward, while leaving in one year of 2020-2023. For newcomers, I examine workers not employed at a treated establishment in 2019 but present in

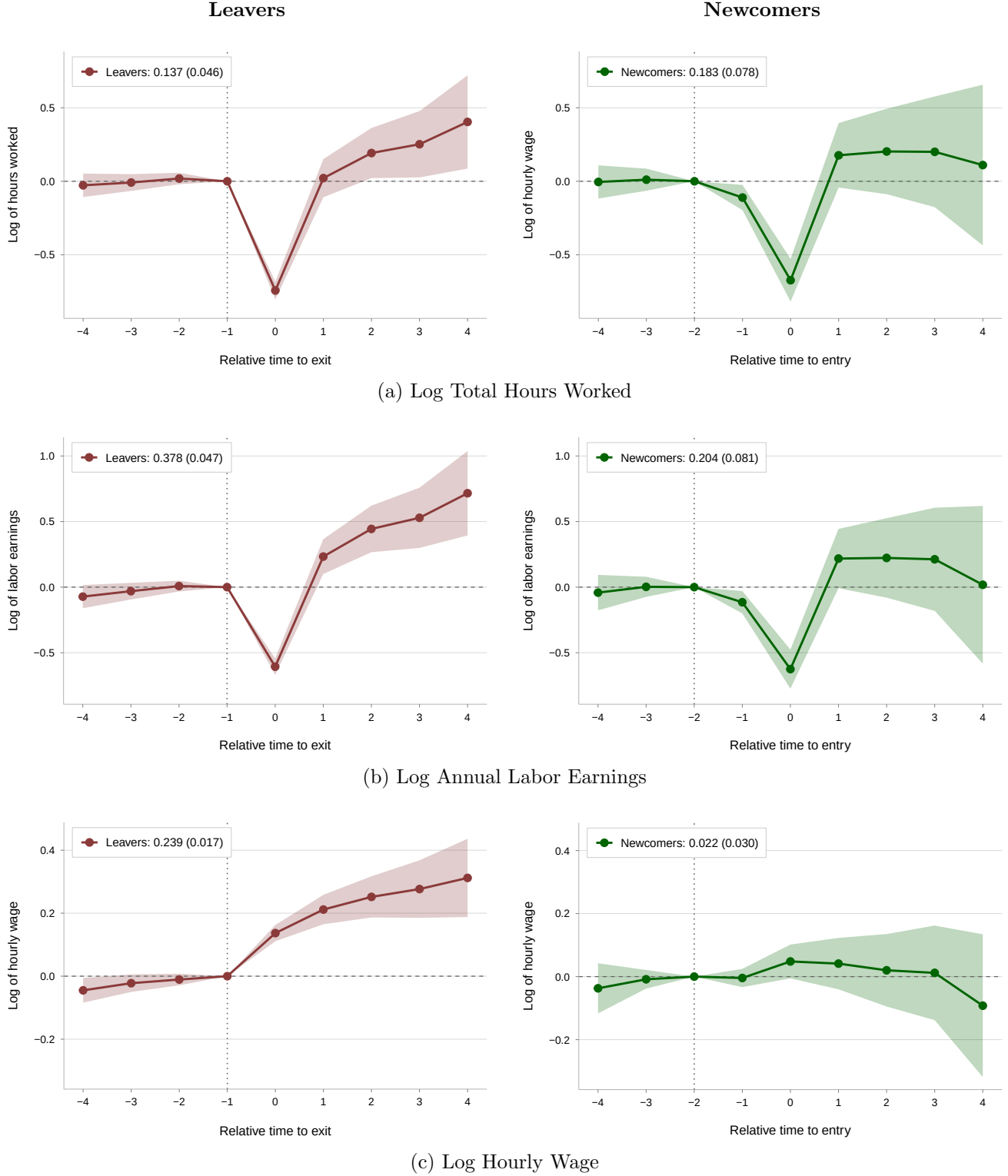
2023, having entered between 2020 and 2023. As a result, treated samples comprises 2,523 leavers and 1,529 newcomers matched to control groups using the same matching procedure detailed in section 3.2.2. It yields 36,576 matched control workers for leavers and 27,131 for newcomers. I estimate relative-time event studies separately for each group as in equation 5. The reference period is $k_0 = 1$ for leavers (the year before separation) and $k_0 = 2$ for newcomers (two years before entry, chosen to capture the pre-exit baseline since workers entering at $t = 0$ likely separated from their previous employer at $t = 1$).

Figure 9 presents the results. For leavers (left column), outcomes exhibit a clear V-shaped pattern. Hours worked drop sharply by 80 percent in the separation year ($t = 0$), but recover fully by the following year ($t = 1$) where the point estimate returns to zero. In subsequent years, hours worked rise substantially, reaching approximately 20 percent above pre-separation levels by $t = 2$ and $t = 3$, following an upward trajectory similar to that observed for stayers. Labor earnings follow a parallel pattern: a 60 percent decline at separation, followed by rapid recovery that exceeds pre-displacement earnings as early as the first post-displacement year. The average treatment effect across all post-displacement years reaches 35 percent in earnings and 14 percent in hours worked, reflecting both longer working time and a substantial hourly wage premium. Most strikingly, hourly wages increase sharply following displacement, rising by more than 20 percent on average across post-displacement years. This wage premium, combined with increased hours, drives the substantial earnings gains.

For newcomers (right column), the dynamics differ notably. Hours worked drop by 70 percent at entry ($t = 0$), consistent with the interpretation that entry often follows separation from a previous employer at $t = 1$. Following entry, hours stabilize at around 15 percent above matched controls, and remains relatively constant across post-entry years, rather than exhibiting the rising trend observed for leavers. However, none of the point estimates is significant at any conventional level. Labor earnings show a similar pattern, with non-significant gains of around 20 percent in $t = 1$ that remain stable or decline slightly in subsequent years. Crucially, hourly wages for newcomers remain flat throughout the post-entry period, though estimates are imprecise as expected from the already non-significant results in both hours and earnings.

These patterns reveal rapid adjustment with minimal persistent displacement costs. Both leavers and newcomers experience sharp but temporary disruptions at the time of their mobility event ($t = 0$), but recover swiftly. Leavers not only recover their pre-separation employment levels

Figure 9: Labor Market Dynamics for Movers in Event Time



Notes: Event-study estimates in relative time around mobility events. Left column: leavers ($N = 2,523$ treated, 36,576 matched controls). Right column: newcomers ($N = 1,529$ treated, 27,131 matched controls). Matched controls selected using the matching procedure detailed in section 3.2.2. Specifications include worker fixed effects, establishment fixed effects, and employment zone-by-year fixed effects. Shaded areas represent 95% confidence intervals based on standard errors clustered at the establishment level. Post-treatment ATTs in legend are precision-weighted averages over 2019-2023.

but substantially exceed them, benefiting from both increased hours and higher hourly wages. The wage premium suggests that skills acquired at establishments adopting heat pump certification are valued in the labor market by other firms (i.e., non firm-specific human capital), enabling workers to secure improved matches following separation. Newcomers, while not experiencing wage gains, benefit from expanded employment opportunities and higher labor earnings through increased hours. The contrast between rising wages for leavers and flat wages for newcomers suggests that experience at establishments engaged in heat pump installation confers valuable skills, while newcomers may be hired for their general labor capacity and trained on the job.

5.3 Heterogeneity Analysis

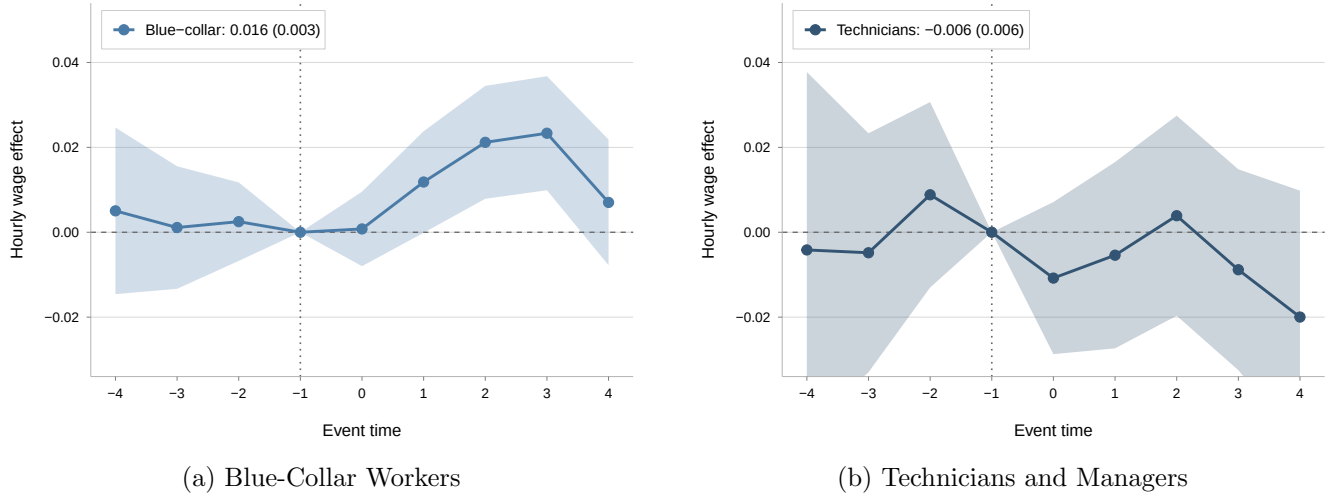
The average effects documented in section 5.2 aggregate over workers with different characteristics and career trajectories. In this section, I examine heterogeneous treatment effects along three dimensions: occupational skill level for stayers, destination establishment’s heat pump certification status for leavers, and industry origin for newcomers.

I decompose the wage effects for stayers by occupational category using the French occupational classification system (PCS), distinguishing between blue-collar workers (primarily trained plumbers, heating technicians, and electricians) and higher-skilled technicians and managers¹⁴. For workers who separate from adopting establishments, I examine whether wage outcomes differ depending on whether they transition to heat pump-certified versus non-certified establishments. For newcomers, I distinguish between workers arriving from HVAC-related establishments (heating, air conditioning, ventilation) versus those originating from non-HVAC sectors. For each analysis, I estimate the relevant specification (equation 2 for stayers, equation 5 for leavers and newcomers) on the subsample of treated workers matched to their respective control groups following the procedure detailed in section 3.2.2.

Figure 10 presents the results for hourly wages across the two groups of stayers. Blue-collar stayers experience a significant hourly wage increase of 1.6 percent on average across post-treatment years (panel 10a), while technicians and managers show a non-significant decline of 0.6 percent (panel 10b). The average +1 percent hourly wage effect for stayers documented in Figure 8 is therefore driven entirely by blue-collar workers.

¹⁴The first level of the French occupational classification system comprises 6 socio-economic groups. The heterogeneity compares group (6) *Ouvriers* versus groups (3) *Cadres et Professions Intellectuelles Supérieures* & (4) *Professions Intermédiaires*.

Figure 10: Heterogeneous Effects on Hourly Wages by Occupation: Stayers



Notes: Event-study estimates of the effect of heat pump certification on log hourly wages for stayers, decomposed by occupational category. Sample restricted to workers employed at the same establishment in 2018, 2019, and 2023. Blue-collar workers ($N = 33,238$ treated, matched controls) comprise trained plumbers, heating technicians, and electricians. Technicians and managers ($N = 9,404$ treated, matched controls) include higher-skilled technical and managerial positions. Each group matched to separate control workers at never-treated establishments following the procedure detailed in section 3.2.2. Specification follows equation 2. Shaded areas represent 95% confidence intervals based on standard errors clustered at the establishment level. Post-treatment ATTs in legends are precision-weighted averages over 2019-2023.

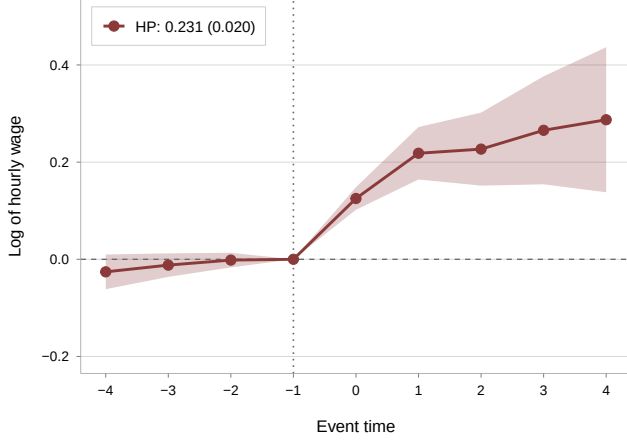
Leavers experience similar wage premiums of approximately 20 percent regardless of whether they transition to heat pump-certified establishments (panel 11a) or non-certified establishments (panel 11b), revealing broad transferability of skills across the HVAC sector. Workers arriving from HVAC-related establishments experience a significant wage premium of 7 percent on average across post-entry years (panel 11c), while those originating from non-HVAC sectors experience a non-significant wage decline (panel 11d). The flat average wage effect for newcomers thus masks substantial heterogeneity by workers' origin, with skill distance governing adjustment costs.

6 Discussion

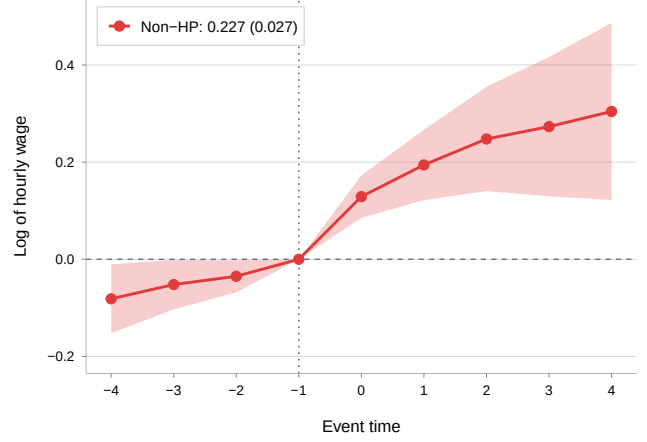
The findings in 5 document substantial labor market reallocation following clean energy technology adoption, yet without imposing persistent costs on displaced workers. Stayers benefit from expanded hours and modest wage gains; separating workers recover swiftly and secure substantial wage premiums; newly hired workers gain employment opportunities and higher earnings. These findings challenge pessimistic narratives about environmental transitions and suggest adaptation within existing SMEs may prove less costly than fossil fuel sector phase-outs dominating just

Figure 11: Heterogeneous Effects on Hourly Wages: Leavers and Newcomers

Leavers by Certification Status at Destination

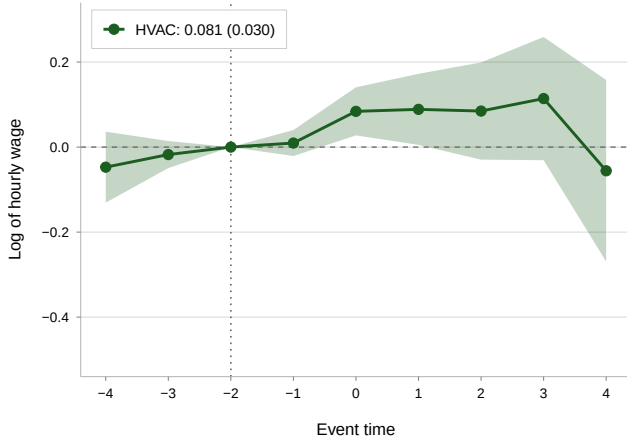


(a) Destination: HP-Certified

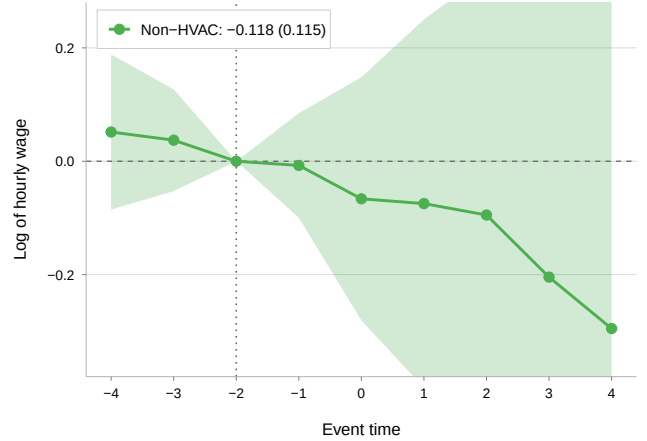


(b) Destination: Non-Certified

Newcomers by Industry Origin



(c) Origin: HVAC Industry



(d) Origin: Non-HVAC Industries

Notes: Event-study estimates in relative time around mobility events. Panels (a)-(b): Leavers present 2016-2019 who separated by 2023, decomposed by destination establishment type. HP-certified destinations ($N = 13,425$ treated, matched controls) hold heat pump RGE certification at worker entry. Non-certified destinations ($N = 3,538$ treated, matched controls) lack heat pump certification. Panels (c)-(d): Newcomers absent in 2019 but employed in 2023 (entry 2020-2023), decomposed by industry of origin. HVAC origin ($N = 19,494$ treated, matched controls) includes prior employment in heating, air conditioning, or ventilation. Non-HVAC origin ($N = 2,912$ treated, matched controls) includes all other sectors. Each group matched to separate control workers at never-treated establishments following the procedure in section 3.2.2. Specification follows equation 5. Shaded areas represent 95% confidence intervals based on standard errors clustered at the establishment level. Post-treatment ATTs in legends are precision-weighted averages over post-mobility years.

transition debates.

6.1 Mechanisms

Two complementary channels explain the observed patterns of labor market adjustment. The first operates through *within-firm reskilling*: stayers and newcomers acquire heat pump competencies on the job, adjusting primarily through expanded hours rather than wage increases. The second operates through *portable skill acquisition and employer mobility*: workers who separate from adopting establishments realize substantial wage premiums because the skills they have accumulated are valued across the HVAC labor market, not merely within their original employer. Together these two channels account for why adjustment costs are low in the aggregate despite significant labor reallocation.

6.1.1 Within-Firm Reskilling

Stayers increase hours by 20 percent immediately post-adoption with only 1 percent wage gains, an adjustment along the intensive margin consistent with on-the-job learning. Newcomers display the same pattern: substantially more hours at flat wages, suggesting they are hired for general labor capacity and trained on the job. These patterns align with the task-based framework (Autor, Levy, and Murnane 2003; Acemoglu and Autor 2011) showing that technology reshapes labor demand by altering job skill content. Establishments facilitate adaptation through on-the-job learning, substantially reducing displacement costs.

Skill distance, defined as the gap between existing and required competencies, determines how costly and rewarding this reskilling proves. Consistent with the task-similarity framework (Gathmann and Schönberg 2010), newcomers from HVAC establishments earn a significant 8.1 percent wage premium (Figure 11c), while those from non-HVAC sectors face flat or slightly negative wage effects (Figure 11d). Among stayers, only blue-collar workers capture a wage gain of 1.6 percent (Figure 10a). By acquiring novel electrical, refrigerant, and thermodynamic competencies, they expand their task range and improve their within-firm bargaining position. Technicians and managers, who are less exposed to the new heat pump requirements, show no significant wage response (Figure 10b).

6.1.2 Portable Skills and Employer Mobility

The sharp contrast in wage outcomes across stayers and leavers reveals a canonical job-ladder pattern, whereby leavers signal their newly acquired skills to prospective employers (Topel and Ward 1992; Postel-Vinay and Robin 2002). Accordingly, despite both groups accumulating similar heat pump experience, workers who change employers secure premiums over 20 percent upon separation, rising thereafter. The gap between the within-firm and market returns reflects wage compression by incumbent employers, who retain trained workers at wages below their market value. This is consistent with monopsony power over a workforce with limited outside options in rural commuting zones.

Heat pump competencies transfer broadly across HVAC applications. Electrical systems, refrigerant handling, complex diagnostics, and thermodynamic principles are relevant to air-to-air heat pumps, ventilation systems, and other high-efficiency installations. The portability of these skills is confirmed directly by the heterogeneity results: leavers earn similar 20 percent wage premiums regardless of whether their destination establishment holds heat pump certification or not (Figures 11a–11b). Skills acquired under the subsidy-driven certification regime thus spill over beyond the subsidized market segment.

6.2 Implications for the Design of a Cost-Minimizing Transition

These mechanisms allow me to formulate two recommendations for policymakers seeking to minimize worker adjustment costs while supporting decarbonization.

6.2.1 A Joint Condition: Market Incentives & Training Supply

The first priority is to create strong market incentives enabling technology adoption within incumbent establishments rather than through creative destruction. On average, workers retain employment relationships and benefit from expanded hours, establishments retain institutional knowledge, and communities avoid concentrated job losses from closures. This contrast sharply with research on fossil fuel industry declines, where workers displaced from coal mining or oil extraction experience substantial and persistent earnings losses (Walker 2013; Haywood, Janser, and Koch 2021; Ellingsen and Espegren 2022; Rud et al. 2024). Similarly, hard-to-abate sectors face fundamental contraction as production declines, with job losses that cannot be offset by comparable opportunities within the same firms—or even local labor market. Policymakers should

therefore set up demand-side subsidies supporting clean technology markets, making adoption attractive for incumbent establishments and facilitating within-firm transitions to minimize worker displacement.

A necessary complement is the access to a rigorous and market-relevant training infrastructure. While heating technicians can easily transfer core competencies in system installation and customer interaction, they need to be trained for electrical and refrigerant-handling skills and advanced thermodynamics. The French case illustrates this joint condition: by 2019, the RGE certification network possessed substantial geographical coverage, with training centers offering standardized courses at costs between €500 and €1,500 per establishment, and distance learning expanding from 2020 onward. This established infrastructure absorbed the demand surge and enabled rapid supply-side response (Figure 2).

6.2.2 Address the Monopsony Bottleneck in Training Investment

A more concerning implication of these findings regards establishments' investment behavior in workers training. Monopsonitic employers compress wages, creating incentives for workers to capture wage premium through separation. Employers who foresee this pattern face reduced incentives to fund training in the first place—especially for portable, non-firm specific skills.

Policymakers should therefore consider complementary interventions addressing this bottleneck. Skill-contingent pay scales in collective agreements, indexing wages to certified competency levels, could reduce the gap between within-firm and market wages and make training more attractive to both parties. Alternatively, a more stringent but publicly subsidized certification process would simultaneously lower entry barriers for new, quality-certified firms, reducing market concentration and the ability of incumbents to capture subsidies, and improve workers skills, directly increasing their bargaining power.

6.3 Limitations and Future Research

Several limitations merit consideration. First, I cannot distinguish voluntary quits from involuntary layoffs. While average leaver outcomes are positive, this may mask heterogeneity: some displaced workers may have experienced persistent losses hidden in aggregate effects. Future research should leverage administrative data on separation reasons to examine self-selected versus forced movers, clarifying whether technology adoption creates genuine losers.

Second, detailed origin-destination matrices would illuminate worker flows. Do transitions occur within narrow sectoral boundaries or across broader HVAC applications? Do newcomers originate from related industries or diverse sectors? These patterns matter for policy: narrow flows suggest binding supply constraints as adoption scales; broad sourcing indicates larger adjustment capacity and wider skill investment benefits.

Third, installation-level data on time spent per heating system would allow a direct test of whether heat pump adoption is labor-augmenting or labor-saving. The hours results documented here—stayers and newcomers working longer without immediate wage gains—are consistent with both interpretations: longer hours may reflect a genuine learning-by-doing investment in a more complex technology, or they may reflect that heat pump installation is inherently more time-intensive than gas boiler installation at comparable productivity. Distinguishing these mechanisms requires observing output per worker-hour at the installation level, which would also clarify whether and when productivity returns to pre-adoption levels as competencies mature.

Fourth, firm-level data on profits and investment would allow an assessment of how the labor cost share evolves following adoption. If heat pump installation is more labor-intensive than fossil boiler work and wages eventually adjust to reflect the new productivity, the share of revenues going to labor may rise, with implications for the long-run sustainability of adoption for small establishments operating on competitive installation margins. This would also shed light on whether the within-firm wage compression identified here represents a temporary transitional phenomenon or a persistent feature of the employer-worker relationship in this sector.

Fifth, the 2019-2023 window captures only short-run dynamics. Extending the time horizon would test whether early-adopter advantages persist as SME electrification advances, or whether initial gains fade as technologies mature and standardize. Long-run analysis could reveal dynamics invisible in this study’s window.

Finally, generalization requires careful consideration. Comparative analysis across countries at different stages of decarbonization, or across sectors with varying skill distances, would establish boundary conditions for the mechanisms identified here, strengthening the evidence base for designing just transition policies.

7 Conclusion

This paper examines labor market outcomes following clean energy technology adoption in France’s heating services industry, a sector emblematic of the SME-level electrification required to achieve net-zero emissions targets. Heat pump certification triggers substantial labor reallocation—simultaneous job creation and destruction—yet workers experience positive average outcomes through within-firm reskilling preserving employment relationships and portable skill acquisition enabling job mobility with wage premiums. Adjustment costs vary with skill distance: workers building on partially transferable competencies secure better outcomes.

The French case offers cautious optimism. Unlike declining fossil sectors imposing persistent displacement costs on workers, the adoption of clean energy technology can preserve existing jobs and create new opportunities for blue-collar workers. Three main policy recommendations emerge from this analysis. First, market incentives such as households subsidies should encourage the technology-adoption by incumbent establishments. Second, manageable skill-distance transitions with accessible training infrastructure should be prioritized. Third, addressing monopsony power is required to alleviate the risk of a training bottleneck, as the green transition scales up.

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A Energy Efficiency Policy in France

A.1 Geographic Distribution of Fuel Oil Heating

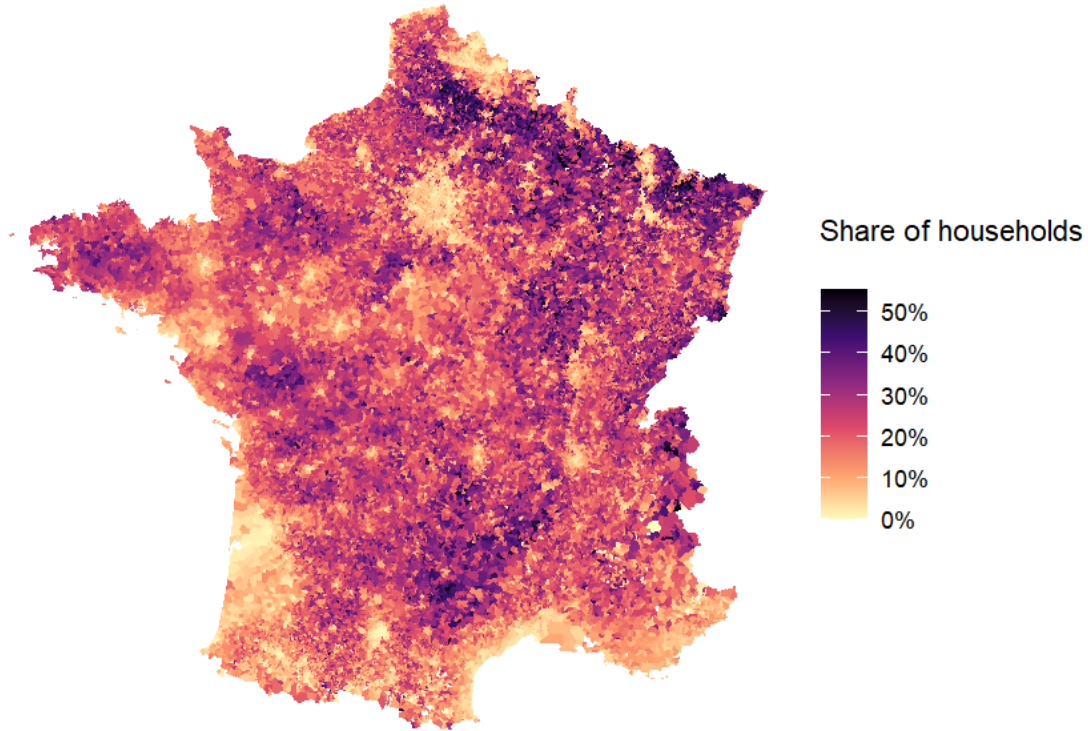


Figure 12: Reliance on a Fuel Oil as Main Heating Source by Municipality, 2017

Notes: The map displays the percentage of primary residences using fuel oil as their main heating source across French municipalities in 2017. Darker shading indicates higher reliance on fuel oil heating. Data from the French Statistical Office (INSEE).

Fuel oil heating is mostly used in detached houses located in rural areas, for which a heat pump represents the main alternative. While the April 2018 reform specifically targeted low-income households, the January 2019 revision granted eligibility to all French households. Figure 12 shows heterogeneity in fuel oil heating across French municipalities in 2017, highlighting the differential incentives created by the policy installers in rural versus urban areas.

A.2 RGE Certification by Technology Category

Figure 13 presents the distribution of RGE (Reconnu Garant de l'Environnement) certificates across different technology categories and certification providers. Heat pump certifications represent a significant share of the total, reflecting both the policy priority assigned to this technology and the substantial training infrastructure developed by certification bodies. The figure also illustrates the diversity of energy efficiency technologies covered by the RGE system, including insulation, efficient gas boilers, biomass heating systems, and photovoltaic installations. Each technology category requires separate certification, creating distinct skill pathways within the broader energy efficiency sector.



Figure 13: RGE Certificates by Technology Category and Certification Provider

Notes: The figure shows the distribution of active RGE certificates across technology categories (heat pumps, insulation, efficient gas boilers, biomass systems, photovoltaic installations, and others) and the main certification providers. Each establishment must obtain a separate certificate for each technology category in which they seek to perform subsidized work. Data from ADEME (French Environment and Energy Management Agency).

A.3 RGE Certification Process

The RGE certification follows a structured process designed to ensure quality standards for subsidized energy efficiency work. Figure 14 illustrates the key steps and timeline.

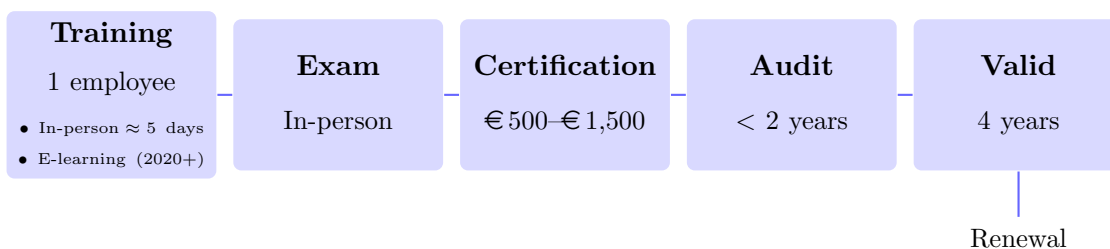


Figure 14: RGE Certification Process Timeline

Notes: Training consists of 21 hours delivered in-person (typically 2-3 consecutive days) or via distance learning. The final QCM exam must be completed in-person regardless of training format (Direction de l’Habitat, de l’Urbanisme et des Paysages 2025). Certification costs approximately €500 to €1,500 per establishment and covers a specific technology category. Certificates remain valid for four years.

Distance Learning Options. While distance learning was technically permitted under the 2014 regulatory framework (Ministère de l’Écologie, du Développement durable et de l’Énergie 2014), it expanded substantially starting in 2020, accelerated by the COVID-19 pandemic (FEEBAT 2020). E-learning can be delivered asynchronously (self-paced modules accessible 24/7) or synchronously (live virtual classrooms) (Elysia Formation 2022; FEEBAT 2025). Training costs for e-learning formats range from €700 to €900, comparable to traditional in-person options (Rénovation et Travaux 2023; Promee 2024). However, distance learning provides substantial indirect cost savings by eliminating travel and accommodation expenses, and reduces business disruption by allowing employees to complete training without extended absences from ongoing projects (Sonergia 2025). The asynchronous format enables learners to progress at their own pace through interactive content including videos, quizzes, and case studies, potentially improving knowledge retention.

A.4 Certification Coverage of Heat Pump Renovation Market

This appendix documents the calculation of coverage ratios for heat pump installations by certified installers over 2016-2018, supporting the argument in Section 2.2.2 that pre-2019 renovation installations occurred nearly universally through certified contractors.

A.4.1 Industry data collection methodology

Market installation statistics reported by Observ’ER 2020 are based on manufacturer and distributor surveys tracking units sold to the professional installer channel. These figures represent

equipment sold to professional installers and wholesalers (2-50 kW range, residential and small commercial), excluding retail/DIY sales. Industry statistics assume minimal inventory lags between manufacturer sales and end-user installations, treating annual sales as a reasonable proxy for annual installations in the same period.

Market composition between new construction and renovation is not systematically tracked in manufacturer sales data, as distributors do not report end-use destination when purchasing equipment. Industry associations provide estimates of renovation versus new construction shares based on installer surveys and market analysis conducted through annual qualitative studies.

A.4.2 Tax credit and EEOs coverage calculation

Table 2 presents the calculation of Tax credit (CITE) and EEO grants beneficiaries and estimate coverage of the renovation market. Tax credit spending data come from official tax authority statistics (Loiseau 2023). Over 2016-2018, tax credit beneficiaries numbered 23,600-38,100 households annually, while EEO grants supported 4,750—9,245 installations (CASD 2024). Crucially, both the tax credit and EEO grants required RGE-certified contractors. Households could in theory cumulate the two sources as of mid-2016, yielding a coverage ratio of 55-81%. Overlap likely remained low given the lack of coordination between obligated actors—proposing EEO grants—and the Public Finances Directorate General (DGFIP) in charge of the tax credit, leading to reasonable coverage of at least two-thirds. This finding demonstrates substantial subsidy penetration of the renovation market. Non-subsidized works (the remaining one third) could also be performed by RGE-certified establishments when administrative burden was judged too high compared to subsidy benefits (e.g., for households in the top income brackets). This proves certification was economically necessary for market participation, making it a reliable proxy for technology adoption.

Table 2: CITE + CEE Coverage of Heat Pump Renovation Market, 2016-2018

Year	CITE (€millions)	CITE (instal.)	CEE (instal.)	Total Installations	Renovation share (%)	Renovation (instal.)	CITE + CEE coverage ratio (%)
2016	212	23,600	4,750	74,475	47	35,003	67—81
2017	259	28,800	5,528	81,700	53	43,301	66—79
2018	343	38,100	9,245	93,580	74	69,225	55—68

Sources: CITE spending from Loiseau 2023, Table 1, p.5. Average CITE amount (€900) calculated as 30% of €3,000 equipment cap per BOFIP 2019. Total air/water sales and renovation share (%) from Observ'ER 2020.

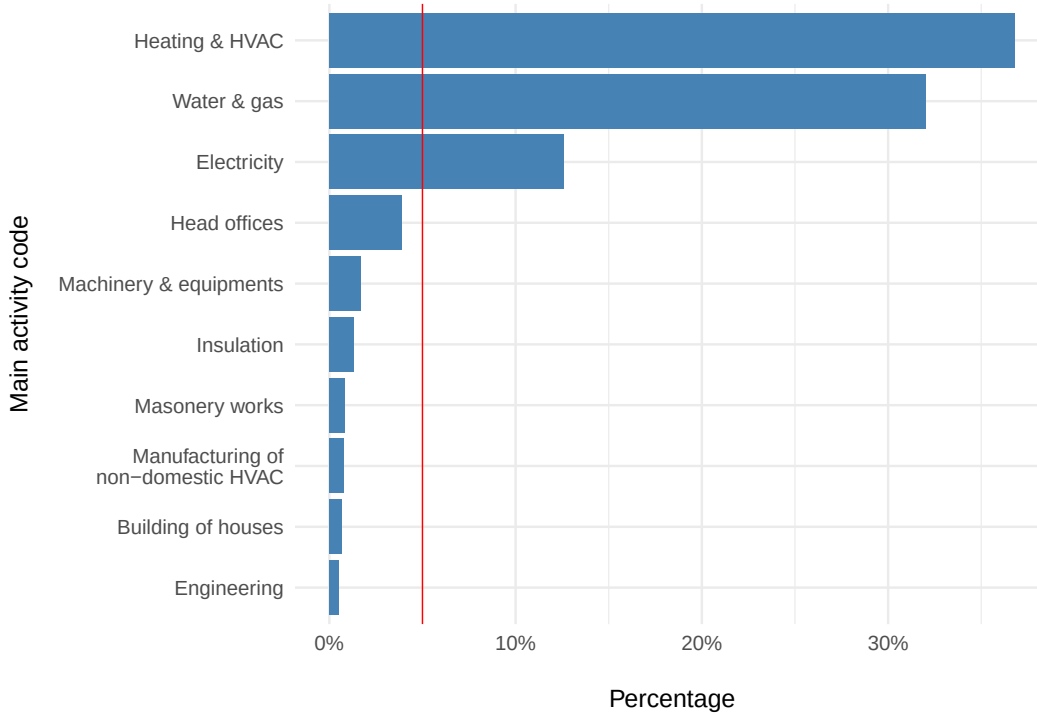
B Data sources

B.1 Worker-level data

I use the *Base Tous Salariés (BTS)*, formerly known as *Déclarations Annuelles de Données Sociales (DADS)*, an exhaustive administrative dataset of French wage earners compiled from employer tax declarations. A key methodological challenge is that the raw data assign each individual a new pseudonymous identifier each year, preventing direct longitudinal tracking. To overcome this limitation, Babet, Godechot, and Palladino 2025 developed a matching algorithm that exploits the overlapping structure of the annual files: each year’s file contains data for both the current and previous year. They match individuals across consecutive years using a combination of stable characteristics including establishment identifier, gender, hours worked, job duration, start and end dates, earnings, age, and municipality of work and residence. This procedure achieves approximately 98% successful matching for the 2002-2023 period, creating what the authors term a “wide panel”—a quasi-exhaustive pseudo-panel that dramatically improves upon the traditional narrow panel (1/12th sample) previously used for French labor market research. The matching algorithm is publicly available and documented in their Appendix C. This enhanced dataset enables more precise estimation of worker and firm fixed effects in the Abowd-Kramarz-Margolis (AKM) framework by including the full universe of mobile workers across firms.

B.2 Sample construction

Figure 15: Top 10 main activity codes across heat pumps certified establishments



B.3 Worker-Level Balance

Table 3 presents detailed worker-level characteristics in 2018 before and after matching. I report age, share of female workers, share of blue-collar workers (primarily trained plumbers, heating technicians, and electricians) and of higher-skilled technicians and managers¹⁵. Exact matching on gender and occupational categories means aggregate balance is not expected on these dimensions—sample-level differences are addressed through the exact matching itself rather than through co-variate balance. Thus, important imbalances persist: female share (15.0% vs. 13.3%), blue-collar workers (67.0% vs. 62.0%), and higher-skilled technicians and managers (17.8% vs. 23.4%). Outcome variables converge: the hourly wage gap narrows from €1.53 to €1.01 and annual earnings converge from €1,290 to €547, while the hours worked gap rises slightly from 54 to 60 hours.

¹⁵The two groups build on the first level of the French occupational classification system (6 socio-economic groups). Blue-collar workers are (6) *Ouvriers*; Higher-skilled technicians and managers are (3) *Cadres et Professions Intellectuelles Supérieures* & (4) *Professions Intermédiaires*.

Table 3: Worker-Level Balance

	Treated		Control		Difference
	Mean (1)	SD (2)	Mean (3)	SD (4)	(T-C) (5)
<i>Panel A: Full Sample (2018)</i>					
N Workers	19,921		479,762		
Age (years)	36.59	(12.63)	37.30	(12.31)	−0.71
Female (%)	15.69	(36.37)	13.33	(33.99)	2.36
Blue collar (%)	64.44	(47.87)	53.68	(49.86)	10.76
Managers (%)	20.10	(40.07)	32.84	(46.96)	−12.74
Hours worked	1,216.19	(670.40)	1,161.97	(687.28)	54.22
Annual earnings (€)	20,117.23	(16,786.53)	21,407.63	(19,241.84)	−1,290.40
Hourly wage (€)	15.79	(8.85)	17.32	(11.06)	−1.53
<i>Panel B: Matched Sample (2018)</i>					
N Workers	13,499		121,681		
Age (years)	36.71	(12.42)	36.84	(12.09)	−0.13
Female (%)	14.97	(35.68)	13.28	(33.94)	1.69
Blue collar (%)	66.97	(47.03)	62.02	(48.53)	4.95
Managers (%)	17.78	(38.24)	23.39	(42.33)	−5.61
Hours worked	1,348.99	(627.43)	1,289.09	(649.36)	59.90
Annual earnings (€)	21,800.65	(15,509.55)	22,347.42	(17,613.84)	−546.77
Hourly wage (€)	15.45	(8.01)	16.46	(8.83)	−1.01

Notes: The table shows worker-level characteristics in 2018 before and after matching. Panel A presents the full sample of all full-time workers (employed at least 30 hours per week) in heating service establishments. Treatment is defined as being employed at an establishment that adopts heat pump technology in 2019. Panel B shows the matched sample using exact matching on establishment activity code (APE), socio-professional category (CSP), and gender, followed by 1:20 nearest-neighbor matching on age. Standard deviations in parentheses.

C Empirical appendix

C.1 Standard Error Calculation for Calendar-Time Aggregation

The calendar-time coefficients γ_t are linear combinations of the cohort-specific event-study coefficients β_k^c . To compute standard errors for γ_t , I use the delta method, which accounts for the covariance structure across the underlying coefficients (Greene 2018).

Let $\boldsymbol{\beta}$ denote the vector of all estimated cohort-specific coefficients $\{\beta_k^c\}$, with corresponding variance-covariance matrix $\mathbf{V}$ obtained from equation (3). Each calendar-time coefficient can be expressed as:

$$\gamma_t = \mathbf{w}_t' \boldsymbol{\beta} \quad (6)$$

where $\mathbf{w}_t$ is a weight vector with elements:

$$w_t^{(c,k)} = \begin{cases} \frac{1}{N_t} & \text{if } c + k = t \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

and $N_t = \sum_{c,k} \mathbb{1}\{c + k = t\}$ is the number of cohort-relative time pairs contributing to calendar year t .

By the delta method, the variance of γ_t is:

$$\text{Var}(\gamma_t) = \mathbf{w}_t' \mathbf{V} \mathbf{w}_t \quad (8)$$

Expanding this expression yields:

$$\text{Var}(\gamma_t) = \frac{1}{N_t^2} \sum_{\substack{c,k: \\ c+k=t}} \sum_{\substack{c',k': \\ c'+k'=t}} \text{Cov}(\beta_k^c, \beta_{k'}^{c'}) \quad (9)$$

The standard error of γ_t is then:

$$\text{SE}(\gamma_t) = \sqrt{\text{Var}(\gamma_t)} \quad (10)$$

This calculation properly accounts for two sources of correlation: (1) correlation between different relative-time coefficients within the same cohort, which arises from shared individual fixed

effects and common shocks affecting that cohort; and (2) correlation between coefficients from different cohorts that contribute to the same calendar year, which can occur due to overlapping observations or common calendar-time shocks captured in the residuals.

For inference, I construct confidence intervals using:

$$\text{CI}_{1-\alpha}(\gamma_t) = \gamma_t \pm z_{\alpha/2} \times \text{SE}(\gamma_t) \quad (11)$$

where $z_{\alpha/2}$ is the appropriate critical value from the standard normal distribution.

D Estimation Results Tables

D.1 Effects on Establishments Technology Adoption

Table 4: Staggered DiD Estimates: Share of Heat Pump Works Following Adoption

Event Time	Estimate	Std. Error
<i>Pre-treatment period</i>		
$t = -8$	-2.3792	(2.9189)
$t = -7$	2.7192	(2.8862)
$t = -6$	-1.5274	(1.7909)
$t = -5$	-0.8614	(1.2233)
$t = -4$	2.5487	(1.1675)
$t = -3$	1.4217	(1.2308)
$t = -2$	6.6828***	(1.2196)
$t = -1$	2.8820	(1.2145)
<i>Post-treatment period</i>		
$t = 0$	21.2409***	(1.3505)
$t = 1$	27.0189***	(1.3857)
$t = 2$	25.1978***	(1.3602)
$t = 3$	24.8695***	(1.4273)
$t = 4$	21.2535***	(1.5887)
$t = 5$	22.6197***	(1.6228)
$t = 6$	24.8158***	(1.9351)
$t = 7$	28.1913***	(2.0247)
$t = 8$	28.2865***	(2.4608)
Overall ATT (post-treatment): 24.8327*** (1.5544)		
Treated establishments	2,859	
Control establishments	92,559	
Estimation method	Doubly Robust	

Notes: This table presents the regression estimates underlying Figure 4a. Staggered difference-in-differences estimates using Callaway and Sant’Anna 2021 methodology. Dependent variable is the share of projected energy savings attributed to air-to-water heat pump installations over total projected savings across all subsidized works performed by the establishment (percentage points). Event time measured in quarters relative to heat pump RGE certification. Estimation uses doubly robust method with never-treated control group. Standard errors clustered at establishment level in parentheses. Overall ATT is the aggregated average treatment effect across all post-treatment periods weighted by group size and treatment exposure. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Staggered DiD Estimates: Share of Fossil Fuel Boiler Works Following Heat Pump Adoption

Event Time	Estimate	Std. Error
<i>Pre-treatment period</i>		
$t = -8$	8.0387*	(4.8363)
$t = -7$	-3.0034	(3.7063)
$t = -6$	4.3512	(4.0103)
$t = -5$	-0.5689	(3.7047)
$t = -4$	-3.0633	(3.6031)
$t = -3$	-6.5573**	(3.0490)
$t = -2$	-6.6273***	(2.0901)
$t = -1$	2.2674	(2.1962)
<i>Post-treatment period</i>		
$t = 0$	-14.0692***	(2.0312)
$t = 1$	-18.0876***	(1.9992)
$t = 2$	-17.4191***	(2.2205)
$t = 3$	-16.7134***	(2.3843)
$t = 4$	-13.9189***	(2.7617)
$t = 5$	-15.2409***	(3.0553)
$t = 6$	-17.2172***	(3.2479)
$t = 7$	-18.2457***	(3.8580)
$t = 8$	-18.2306***	(4.2740)
Overall ATT (post-treatment): -16.5714*** (2.8594)		
Treated establishments	2,859	
Control establishments	92,559	
Estimation method	Doubly Robust	

Notes: This table presents the regression estimates underlying Figure 4b. Staggered difference-in-differences estimates using Callaway and Sant’Anna 2021 methodology. Dependent variable is the share of projected energy savings attributed to fossil fuel boiler installations over total projected savings across all subsidized works performed by the establishment (percentage points). Event time measured in quarters relative to heat pump RGE certification. Estimation uses doubly robust method with never-treated control group. Standard errors clustered at establishment level in parentheses. Overall ATT is the aggregated average treatment effect across all post-treatment periods weighted by group size and treatment exposure. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.2 Effects on Establishments Employment Behavior

Table 6: Staggered DiD Estimates: Job Entries Following Heat Pump Adoption

Event Time	Estimate	Std. Error
<i>Pre-treatment period</i>		
$t = -8$	0.0422	(0.0374)
$t = -7$	0.0342	(0.0394)
$t = -6$	0.0148	(0.0351)
$t = -5$	0.0818**	(0.0410)
$t = -4$	0.0782**	(0.0393)
$t = -3$	0.0779*	(0.0404)
$t = -2$	0.0879**	(0.0388)
$t = -1$	0.0871**	(0.0428)
<i>Post-treatment period</i>		
$t = 0$	0.1558***	(0.0385)
$t = 1$	0.3762***	(0.0782)
$t = 2$	0.5550***	(0.1241)
$t = 3$	0.7493***	(0.1703)
$t = 4$	0.9364***	(0.2002)
$t = 5$	1.0735***	(0.2125)
$t = 6$	1.2206***	(0.2551)
$t = 7$	1.3903***	(0.2825)
$t = 8$	1.5936***	(0.3070)
Overall ATT (post-treatment): 0.8945*** (0.1739)		
Treated establishments	2,859	
Control establishments	92,559	
Estimation method	Doubly Robust	

Notes: This table presents the regression estimates underlying Figure 5a. Staggered difference-in-differences estimates using Callaway and Sant’Anna 2021 methodology. Dependent variable is cumulative job entries at the establishment level. Event time measured in quarters relative to heat pump RGE certification. Estimation uses doubly robust method with never-treated control group. Standard errors clustered at establishment level in parentheses. Overall ATT is the aggregated average treatment effect across all post-treatment periods weighted by group size and treatment exposure. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7: Staggered DiD Estimates: Job Exits Following Heat Pump Adoption

Event Time	Estimate	Std. Error
<i>Pre-treatment period</i>		
$t = -8$	0.0682*	(0.0375)
$t = -7$	0.0649**	(0.0279)
$t = -6$	0.0385**	(0.0187)
$t = -5$	0.0409**	(0.0180)
$t = -4$	0.0786***	(0.0232)
$t = -3$	0.0578**	(0.0226)
$t = -2$	0.0471**	(0.0191)
$t = -1$	0.0280	(0.0200)
<i>Post-treatment period</i>		
$t = 0$	0.0945***	(0.0251)
$t = 1$	0.1896***	(0.0451)
$t = 2$	0.2813***	(0.0616)
$t = 3$	0.3382***	(0.0666)
$t = 4$	0.4305***	(0.0865)
$t = 5$	0.5082***	(0.0899)
$t = 6$	0.5865***	(0.1026)
$t = 7$	0.6815***	(0.1114)
$t = 8$	0.8135***	(0.1205)
Overall ATT (post-treatment): 0.4360*** (0.0714)		
Treated establishments	2,859	
Control establishments	92,559	
Estimation method	Doubly Robust	

Notes: This table presents the regression estimates underlying Figure 5b. Staggered difference-in-differences estimates using Callaway and Sant’Anna 2021 methodology. Dependent variable is cumulative job exits at the establishment level. Event time measured in quarters relative to heat pump RGE certification. Estimation uses doubly robust method with never-treated control group. Standard errors clustered at establishment level in parentheses. Overall ATT is the aggregated average treatment effect across all post-treatment periods weighted by group size and treatment exposure. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.3 Workers-level Career Trajectories

D.3.1 Effect on incumbent workers

Table 8: Effect of Heat Pump Certification on Incumbent Workers: Log Total Hours Worked

Event Time	Estimate	Std. Error
<i>Pre-treatment period</i>		
$t = -4$	0.007	(0.017)
$t = -3$	0.011	(0.015)
$t = -2$	0.012	(0.016)
$t = -1$	[Reference period]	
<i>Post-treatment period</i>		
$t = 0$	0.139***	(0.015)
$t = 1$	0.119***	(0.015)
$t = 2$	0.116***	(0.017)
$t = 3$	0.069***	(0.018)
$t = 4$	0.067***	(0.017)
Observations	870,187	
Adjusted R ²	0.3592	
Within R ²	0.0005	

Notes: This table presents the regression estimates underlying Figure 6a. Event-study estimates of the effect of heat pump certification on log total hours worked for workers employed at treated establishments in 2018 and 2019. Event time measured in years relative to certification year (2019). Control workers matched exactly on establishment activity code (APE), socio-professional category (CSP), and gender, followed by 1:20 nearest-neighbor matching on age. Specification includes worker fixed effects, establishment fixed effects (SIREN and NIC), and year-by-commuting zone fixed effects. Standard errors two-way clustered at worker and establishment levels in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9: Effect of Heat Pump Certification on Incumbent Workers: Log Annual Labor Earnings

Event Time	Estimate	Std. Error
<i>Pre-treatment period</i>		
$t = -4$	0.008	(0.018)
$t = -3$	0.014	(0.015)
$t = -2$	0.017	(0.016)
$t = -1$	[Reference period]	
<i>Post-treatment period</i>		
$t = 0$	0.144***	(0.016)
$t = 1$	0.133***	(0.015)
$t = 2$	0.138***	(0.018)
$t = 3$	0.094***	(0.018)
$t = 4$	0.072***	(0.018)
Observations	870,207	
Adjusted R ²	0.5354	
Within R ²	0.0006	

Notes: This table presents the regression estimates underlying Figure 6b. Event-study estimates of the effect of heat pump certification on log annual labor earnings for workers employed at treated establishments in 2018 and 2019. Event time measured in years relative to certification year (2019). Control workers matched exactly on establishment activity code (APE), socio-professional category (CSP), and gender, followed by 1:20 nearest-neighbor matching on age. Specification includes worker fixed effects, establishment fixed effects (SIREN and NIC), and year-by-commuting zone fixed effects. Standard errors two-way clustered at worker and establishment levels in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 10: Effect of Heat Pump Certification on Incumbent Workers: Log Hourly Wage

Event Time	Estimate	Std. Error
<i>Pre-treatment period</i>		
$t = -4$	0.001	(0.007)
$t = -3$	0.003	(0.006)
$t = -2$	0.005	(0.004)
$t = -1$	[Reference period]	
<i>Post-treatment period</i>		
$t = 0$	0.004	(0.003)
$t = 1$	0.014**	(0.005)
$t = 2$	0.022***	(0.005)
$t = 3$	0.025***	(0.006)
$t = 4$	0.006	(0.006)
Observations	870,187	
Adjusted R ²	0.6881	
Within R ²	0.0001	

Notes: This table presents the regression estimates underlying Figure 6c. Event-study estimates of the effect of heat pump certification on log hourly wage for workers employed at treated establishments in 2018 and 2019. Event time measured in years relative to certification year (2019). Control workers matched exactly on establishment activity code (APE), socio-professional category (CSP), and gender, followed by 1:20 nearest-neighbor matching on age. Specification includes worker fixed effects, establishment fixed effects (SIREN and NIC), and year-by-commuting zone fixed effects. Standard errors two-way clustered at worker and establishment levels in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.3.2 Decomposing Effects: Stayers versus leavers

Table 11: Effects on Hours Worked: Stayers vs. Leavers (Calendar Time)

	Stayers		Leavers	
Calendar Year	Estimate	Std. Error	Estimate	Std. Error
<i>Pre-treatment period</i>				
2015	−0.004	(0.021)	−0.008	(0.041)
2016	0.006	(0.019)	0.011	(0.056)
2017	0.006	(0.020)	0.042	(0.056)
2018	[Reference]		0.004	(0.059)
<i>Post-treatment period</i>				
2019	0.191***	(0.018)	−0.302***	(0.084)
2020	0.218***	(0.017)	−0.427***	(0.096)
2021	0.220***	(0.020)	−0.312***	(0.087)
2022	0.204***	(0.018)	−0.279***	(0.066)
2023	0.124***	(0.019)	−0.411***	(0.130)
Observations	739,022		19,352	
Adjusted R ²	0.3616		—	
Within R ²	0.0014		—	

Notes: This table presents the regression estimates underlying Figure 7a. Comparison of treatment effects on log total hours worked for stayers versus leavers, both expressed in calendar time. **Stayers** are workers employed at the same establishment in 2018, 2019, and 2023. Event-study specification with worker fixed effects, establishment fixed effects, and year-by-commuting zone fixed effects. Standard errors clustered at establishment level. **Leavers** are workers present in 2018-2019 but separated by 2023 (minimum 3 years tenure). Estimates are calendar-time aggregations of cohort-specific event-study coefficients from equation (3), averaged across separation cohorts active in each year. Standard errors computed using delta method accounting for covariance across cohorts. Each group matched to separate control workers at never-treated establishments. *** p<0.01, ** p<0.05, * p<0.1.

Table 12: Effects on Annual Earnings: Stayers vs. Leavers (Calendar Time)

	Stayers		Leavers	
Calendar Year	Estimate	Std. Error	Estimate	Std. Error
<i>Pre-treatment period</i>				
2015	−0.003	(0.023)	−0.060	(0.043)
2016	0.009	(0.019)	−0.015	(0.057)
2017	0.010	(0.020)	0.022	(0.060)
2018	[Reference]		0.045	(0.060)
<i>Post-treatment period</i>				
2019	0.190***	(0.018)	−0.236***	(0.086)
2020	0.226***	(0.017)	−0.343***	(0.099)
2021	0.236***	(0.020)	−0.304***	(0.089)
2022	0.218***	(0.019)	−0.229***	(0.068)
2023	0.127***	(0.021)	−0.374***	(0.117)
Observations	739,038		19,352	
Adjusted R ²	0.5334		—	
Within R ²	0.0014		—	

Notes: This table presents the regression estimates underlying Figure 7b. Comparison of treatment effects on log annual labor earnings for stayers versus leavers, both expressed in calendar time. **Stayers** are workers employed at the same establishment in 2018, 2019, and 2023. Event-study specification with worker fixed effects, establishment fixed effects, and year-by-commuting zone fixed effects. Standard errors clustered at establishment level. **Leavers** are workers present in 2018-2019 but separated by 2023 (minimum 3 years tenure). Estimates are calendar-time aggregations of cohort-specific event-study coefficients from equation (3), averaged across separation cohorts active in each year. Standard errors computed using delta method accounting for covariance across cohorts. Each group matched to separate control workers at never-treated establishments. *** p<0.01, ** p<0.05, * p<0.1.

Table 13: Effect of Heat Pump Certification on Hourly Wages: Stayers

Event Time	Estimate	Std. Error
<i>Pre-treatment period</i>		
$t = -4$	0.001	(0.010)
$t = -3$	0.003	(0.007)
$t = -2$	0.004	(0.005)
$t = -1$	[Reference period]	
<i>Post-treatment period</i>		
$t = 0$	-0.002	(0.004)
$t = 1$	0.008	(0.006)
$t = 2$	0.016**	(0.006)
$t = 3$	0.014**	(0.006)
$t = 4$	0.002	(0.007)
Observations	739,022	
Adjusted R ²	0.6860	
Within R ²	3.54×10^{-5}	

Notes: This table presents the regression estimates underlying Figure 8. Event-study estimates of the effect of heat pump certification on log hourly wage for stayers—workers employed at the same establishment in 2018, 2019, and 2023. Event time measured in years relative to certification year (2019). Control workers matched exactly on establishment activity code (APE), socio-professional category (CSP), and gender, followed by nearest-neighbor matching on age. Specification includes worker fixed effects, establishment fixed effects (SIREN and NIC), and year-by-commuting zone fixed effects. Standard errors clustered at establishment level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

D.3.3 Worker Mobility and Adjustment Dynamics

Table 14: Labor Market Dynamics for Leavers in Event Time

	Hours		Earnings		Hourly Wage	
Rel. Time	Est.	Std. Err.	Est.	Std. Err.	Est.	Std. Err.
<i>Pre-separation</i>						
$t = -4$	-0.027	(0.041)	-0.072	(0.045)	-0.045**	(0.020)
$t = -3$	-0.009	(0.029)	-0.031	(0.032)	-0.022	(0.014)
$t = -2$	0.019	(0.020)	0.008	(0.021)	-0.011	(0.009)
$t = -1$	[Reference period]					
<i>Post-separation</i>						
$t = 0$	-0.743***	(0.031)	-0.606***	(0.032)	0.137***	(0.013)
$t = 1$	0.021	(0.066)	0.233***	(0.068)	0.212***	(0.024)
$t = 2$	0.192**	(0.087)	0.444***	(0.091)	0.252***	(0.033)
$t = 3$	0.252**	(0.115)	0.528***	(0.117)	0.277***	(0.047)
$t = 4$	0.404**	(0.162)	0.716***	(0.164)	0.312***	(0.063)
Observations	19,352		19,352		19,352	
Adjusted R ²	0.4466		0.5487		0.7201	
Within R ²	0.1685		0.1395		0.0238	

Notes: This table presents the regression estimates underlying Figure 9 (left column). Event-study estimates in relative time around separation for leavers—workers present at treated establishments 2016-2019 who separated by 2023 (minimum 3 years tenure). Relative time $t = 0$ is the separation year; $t = -1$ (year before separation) is the reference period. Outcomes are log total hours worked, log annual labor earnings, and log hourly wage. Control workers matched exactly on age, socio-professional category (PCS), establishment activity code, and gender. Specification includes worker fixed effects, establishment fixed effects (SIREN and NIC), cohort fixed effects, and year-by-commuting zone fixed effects. Standard errors clustered at establishment level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 15: Labor Market Dynamics for Newcomers in Event Time

Rel. Time	Hours		Earnings		Hourly Wage	
	Est.	Std. Err.	Est.	Std. Err.	Est.	Std. Err.
<i>Pre-entry</i>						
$t = -4$	-0.005	(0.058)	-0.042	(0.069)	-0.037	(0.041)
$t = -3$	0.011	(0.038)	0.002	(0.039)	-0.008	(0.015)
$t = -2$	[Reference period]					
$t = -1$	-0.111**	(0.044)	-0.115**	(0.044)	-0.004	(0.015)
<i>Post-entry</i>						
$t = 0$	-0.673***	(0.074)	-0.625***	(0.076)	0.048*	(0.027)
$t = 1$	0.177	(0.112)	0.218*	(0.115)	0.041	(0.042)
$t = 2$	0.203	(0.148)	0.223	(0.155)	0.020	(0.059)
$t = 3$	0.200	(0.192)	0.212	(0.201)	0.012	(0.077)
$t = 4$	0.110	(0.279)	0.018	(0.307)	-0.092	(0.116)
Observations	13,762		13,762		13,762	
Adjusted R ²	0.3627		0.4646		0.4808	
Within R ²	0.1204		0.0967		0.0017	

Notes: This table presents the regression estimates underlying Figure 9 (right column). Event-study estimates in relative time around entry for newcomers—workers not employed at treated establishments in 2019 but present in 2023 (entry 2020-2023). Relative time $t = 0$ is the entry year; $t = -2$ (two years before entry) is the reference period, chosen to capture the pre-exit baseline since workers entering at $t = 0$ likely separated from their previous employer at $t = -1$. Outcomes are log total hours worked, log annual labor earnings, and log hourly wage. Control workers matched exactly on age, socio-professional category (PCS), establishment activity code, and gender. Specification includes worker fixed effects, establishment fixed effects (SIREN and NIC), and year-by-commuting zone fixed effects. Standard errors two-way clustered at worker and establishment levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D.4 Heterogeneity Analysis

Table 16: Heterogeneous Effects on Hourly Wages by Occupation: Stayers

	Blue-Collar		Technicians & Managers	
Event Time	Estimate	Std. Error	Estimate	Std. Error
<i>Pre-treatment period</i>				
$t = -4$	0.005	(0.010)	-0.004	(0.021)
$t = -3$	0.001	(0.007)	-0.005	(0.014)
$t = -2$	0.002	(0.005)	0.009	(0.011)
$t = -1$	[Reference period]			
<i>Post-treatment period</i>				
$t = 0$	0.001	(0.004)	-0.011	(0.009)
$t = 1$	0.012+	(0.006)	-0.005	(0.011)
$t = 2$	0.021**	(0.007)	0.004	(0.012)
$t = 3$	0.023***	(0.007)	-0.009	(0.012)
$t = 4$	0.007	(0.008)	-0.020	(0.015)
Observations	467,729		176,275	
Adjusted R ²	0.6507		0.6136	
Within R ²	8.01×10^{-5}		3.97×10^{-5}	

Notes: This table presents the regression estimates underlying Figure 10. Event-study estimates of the effect of heat pump certification on log hourly wages for stayers, decomposed by occupational category. Sample restricted to workers employed at the same establishment in 2018, 2019, and 2023. **Blue-collar workers** comprise trained plumbers, heating technicians, and electricians (PCS: *Ouvriers*). **Technicians and managers** include higher-skilled technical and managerial positions (PCS: *Cadres & Professions Intellectuelles Supérieures* and *Professions Intermédiaires*). Each group matched to separate control workers at never-treated establishments following the matching procedure detailed in section 5.2. Specification includes worker fixed effects, establishment fixed effects, and year-by-commuting zone fixed effects. Standard errors two-way clustered at worker and establishment levels. *** p<0.01, ** p<0.05, * p<0.1.

Table 17: Heterogeneous Effects on Hourly Wages by Destination: Leavers

	HP-Certified Dest.		Non-Certified Dest.	
Rel. Time	Estimate	Std. Error	Estimate	Std. Error
<i>Pre-separation</i>				
$t = -4$	-0.026	(0.018)	-0.081*	(0.036)
$t = -3$	-0.012	(0.012)	-0.052*	(0.026)
$t = -2$	-0.002	(0.008)	-0.035*	(0.017)
$t = -1$	[Reference period]			
<i>Post-separation</i>				
$t = 0$	0.125***	(0.012)	0.129***	(0.022)
$t = 1$	0.218***	(0.028)	0.194***	(0.037)
$t = 2$	0.227***	(0.038)	0.248***	(0.055)
$t = 3$	0.265***	(0.057)	0.273***	(0.073)
$t = 4$	0.287***	(0.076)	0.304**	(0.093)
Observations	13,870		5,423	
Adjusted R ²	0.7121		0.6887	
Within R ²	0.0227		0.0209	

Notes: This table presents the regression estimates underlying Figure 11 (panels a-b). Event-study estimates in relative time around separation for leavers, decomposed by destination establishment certification status. Sample: workers present at treated establishments 2016-2019 who separated by 2023 (minimum 3 years tenure). Relative time $t = 0$ is the separation year; $t = -1$ is the reference period. **HP-certified destinations** hold heat pump RGE certification at worker entry. **Non-certified destinations** lack heat pump certification. Each group matched to separate control workers at never-treated establishments. Specification includes worker fixed effects, establishment fixed effects, cohort fixed effects, and year-by-commuting zone fixed effects. Standard errors two-way clustered at worker and establishment levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 18: Heterogeneous Effects on Hourly Wages by Industry Origin: Newcomers

Rel. Time	HVAC Origin		Non-HVAC Origin	
	Estimate	Std. Error	Estimate	Std. Error
<i>Pre-entry</i>				
$t = -4$	-0.047	(0.043)	0.052	(0.070)
$t = -3$	-0.018	(0.016)	0.037	(0.046)
$t = -2$	[Reference period]			
$t = -1$	0.010	(0.016)	-0.007	(0.047)
<i>Post-entry</i>				
$t = 0$	0.084**	(0.029)	-0.066	(0.109)
$t = 1$	0.089*	(0.043)	-0.075	(0.166)
$t = 2$	0.085	(0.058)	-0.095	(0.218)
$t = 3$	0.114	(0.074)	-0.204	(0.298)
$t = 4$	-0.056	(0.109)	-0.295	(0.400)
Observations	12,529		1,364	
Adjusted R ²	0.4921		0.7234	
Within R ²	0.0024		0.0087	

Notes: This table presents the regression estimates underlying Figure 11 (panels c-d). Event-study estimates in relative time around entry for newcomers, decomposed by industry of origin. Sample: workers not employed at treated establishments in 2019 but present in 2023 (entry 2020-2023). Relative time $t = 0$ is the entry year; $t = -2$ is the reference period. **HVAC origin** includes prior employment in heating, air conditioning, or ventilation establishments. **Non-HVAC origin** includes all other sectors. Each group matched to separate control workers at never-treated establishments. Specification includes worker fixed effects, establishment fixed effects, and year-by-commuting zone fixed effects. Standard errors two-way clustered at worker and establishment levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.